Missing Data Imputation using Genetic Algorithms: A Distributional Approach

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Missing Data Imputation using Genetic Algorithms: A Distributional Approach

ABSTRACT

We prove that distributional fidelity and pointwise accuracy are structurally orthogonal objectives in missing data imputation: no single method can simultaneously minimise both. The RMSE-minimising imputation (conditional expectation) must have strictly positive distributional distance F_r by the law of total variance; conversely, any F_r-zero imputation must have positive RMSE unless the missing values follow the same marginal distribution as the observed data. We further show this impossibility extends to neural methods: for GAIN (Generative Adversarial Imputation Networks), no reconstruction weight $\alpha > 0$ allows joint minimisation of F_r and RMSE, establishing C2 as a *universal impossibility theorem* for imputation.

We validate these results through a comprehensive evaluation of MIGA (Missing data Imputation via Genetic Algorithm), which directly optimises F_r — a fitness function penalising discrepancies in means, covariances, and skewness — against MICE, KNN, and GAIN across five UCI benchmark datasets (Iris, Wine, Glass, Haberman, Wholesale) at 30% MAR. MIGA achieves 1.5–2.0× lower F_r than MICE on low-dimensional datasets (Wilcoxon $p < 0.001$), while MICE achieves 1.4–2.1× lower RMSE on every dataset. An empirical α-sweep for GAIN ($\alpha \in \{0, 1, 10, 100, 1000, 10000\}$, all 5 datasets) confirms the impossibility holds across all reconstruction weights, revealing three failure regimes: monotone-but-off-frontier on Iris and Wine, training instability at high α on Haberman, and domination by MICE on both metrics on Glass.

Four additional findings characterise the scope and limits of MIGA's advantage. First, a *dimension–correlation interaction* governs MIGA's advantage: a controlled $p \times \rho$ grid shows MIGA wins for $p \leq 13$ at $\rho \leq 0.6$ but loses at $\rho = 0.9$ even for $p = 8$. Second, *directional MNAR* produces a striking inversion: under top-quantile missingness on Haberman, MIGA simultaneously achieves the global minimum F_r (0.810) and global maximum RMSE (0.384). Third, *Adaptive Diversity Injection* (AD-MIGA) replaces the fixed 90% injection rate with an exponential decay schedule $\alpha(g) = \exp(-\lambda g/G)$, achieving lower F_r on all 3 tested datasets and converging 33 generations earlier on average. Fourth, downstream evaluation shows MIGA's distributional advantage propagates to valid statistical inference on well-conditioned datasets (40–100% KS-test pass rate, 70–95% bootstrap CI coverage).

The thesis delivers nine contributions: the first open-source reimplementation of MIGA; the first formal proof of F_r–RMSE orthogonality; the first MNAR evaluation of MIGA; the first controlled $p \times \rho$ characterisation; propensity-score-weighted IPW-F_r (reduces F_r by 1–38% when $n/p \gtrsim 20$); AD-MIGA adaptive diversity injection; universal orthogonality via GAIN comparison and full α-sweep; and near-linear parallel speedup (1.94–1.98× at Q=2) via `n_jobs`.

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List of Abbreviations

AD-MIGA	**A**daptive **D**iversity injection **MIGA**
CI	**C**onfidence **I**nterval
GA	**G**enetic **A**lgorithm
GAIN	**G**enerative **A**dversarial **I**mputation **N**etworks
F_r	Fréchet-inspired distributional **F**itness function
IPW	**I**nverse **P**robability **W**eighting
KNN	**K**-**N**earest **N**eighbours
KS	**K**olmogorov-**S**mirnov (test)
LW	**L**edoit-**W**olf (covariance shrinkage)
MAR	**M**issing **A**t **R**andom
MCAR	**M**issing **C**ompletely **A**t **R**andom
MICE	**M**ultiple **I**mputation by **C**hained **E**quations
MIGA	**M**issing data **I**mputation via **G**enetic **A**lgorithm
MNAR	**M**issing **N**ot **A**t **R**andom
NRMSE	**N**ormalised **R**oot **M**ean **S**quared **E**rror
RMSE	**R**oot **M**ean **S**quared **E**rror
UCI	**U**niversity of **C**alifornia, **I**rvine (ML Repository)

Chapter 1

Introduction

1.1 Background and Motivation

Missing data is a pervasive challenge in real-world machine learning and statistical analysis. Medical records omit test results for patients who did not present with relevant symptoms; financial surveys suffer from non-response bias; sensor networks produce gaps due to transmission failures or device faults. The proportion of missing values in real-world datasets commonly ranges from 5% to 40% [1], and naive strategies for handling missingness can severely compromise the integrity of downstream analyses.

Three broad strategies are typically employed. **Complete case analysis** discards any observation with at least one missing value, sacrificing statistical power and potentially introducing selection bias. **Single imputation** (e.g., replacing missing values with column means) fills the gaps with a single plausible value, but distorts the variance structure of the data and treats imputed values as though they were observed. **Multiple imputation** generates several plausible complete datasets and combines inferences across them, appropriately accounting for the uncertainty introduced by missingness.

Among multiple imputation methods, MICE (Multiple Imputation by Chained Equations; van Buuren and Groothuis-Oudshoorn, 2011) has become the standard [2]. MICE iteratively regresses each incomplete variable on all other variables, producing imputations that are optimal in the sense of minimising expected squared error. However, this optimality comes at a cost: MICE imputes conditional means, which suppresses marginal variance. Van Buuren (2018, §2.6) explicitly notes that "the minimum mean squared error is achieved by regression imputation, which suppresses variance and produces biased statistical inference" [1]. For any analysis that requires the correct marginal distribution of the imputed variables — distributional testing, bootstrap inference, synthetic data generation — MICE produces systematically distorted results.

This motivates a distributional approach to imputation: rather than minimising pointwise squared error, find imputed values whose *distribution* matches the observed distribution of complete cases. This is precisely the objective of MIGA (Missing data Imputation via Genetic Algorithm).

1.2 The MIGA Algorithm

Figueroa-García, Neruda, and Hernandez-Pérez (2023) proposed MIGA, published in *Information Sciences* [3]. MIGA frames missing data imputation as an optimisation problem: find the set of imputed values that minimises the statistical distance between the set of complete rows (X_A) and the set of rows with at least one missing value after imputation (X_C).

The fitness function (F_r) is a sum of three Minkowski distances measuring differences in mean vectors, covariance structure, and skewness between X_A and X_C:

$$F_r = D_r(\tilde{x}_A, \tilde{x}_C) + D_r(\tilde{S}, I) + D_r(b_A, b_C) \tag{1.1}$$

where $\tilde{x}$ denotes standardised means, $\tilde{S} = S_A^{-1/2} S_C S_A^{-1/2}$ is the relative covariance, and b denotes skewness vectors. A genetic algorithm with l individuals over G generations across Q independent runs is used to minimise F_r.

MIGA handles mixed variable types (continuous, discrete, binary) natively through variable-specific random generators. It imposes no parametric assumptions on the data distribution and explicitly targets distributional fidelity rather than pointwise accuracy, making it the correct tool when downstream analysis requires correct distributional properties.

However, the original paper provides no public implementation, and several implementation details are underspecified. No systematic comparison against modern baselines (KNN, MICE) is reported, and the algorithm has not been evaluated under Missing Not At Random (MNAR) conditions. This thesis addresses all three gaps.

1.3 Research Objectives

This thesis pursues five primary objectives:

1. **Reproduce MIGA** from scratch, providing the first open-source implementation and verifying replication of the paper's results on five benchmark datasets.
2. **Formally prove the F_r–RMSE orthogonality**, establishing that no single imputation can jointly minimise both distributional distance and pointwise error.
3. **Evaluate MIGA under MNAR**, providing the first systematic assessment of MIGA's distributional objective when the missing mechanism is not at random, including an IPW-F_r correction for selection bias.

4. **Characterise when MIGA's F_r advantage holds**, via systematic baseline comparison across five datasets and a controlled $p \times \rho$ synthetic experiment isolating dimensionality and correlation effects.

1.4 Contributions

The specific contributions of this thesis are:

C1 — Open-source reimplementation. A complete Python reimplementation of MIGA from the paper specification, with reproducible experiments across five benchmark datasets (Iris, Wine, Glass, Haberman, Wholesale). Implementation decisions not specified in the paper — generator types, covariance conditioning, stopping criteria — are documented explicitly. This is the first public implementation of MIGA.

C2 — Formal F_r–RMSE orthogonality theorem. We prove that the RMSE-minimising imputation (conditional expectation) must have strictly positive F_r whenever conditional variance is non-zero (Proposition 2.2), and that any F_r-zero imputation must have positive RMSE unless the missing values are drawn from the same distribution as the observed rows (Proposition 2.3). This establishes that no single imputation can jointly minimise both objectives, formalising the empirical decoupling observed across all experiments.

C3 — First MNAR evaluation. We extend MIGA's evaluation to three MNAR mechanisms (top-quantile, bottom-quantile, tails censoring) across five datasets. Under top-quantile MNAR on Haberman, MIGA simultaneously achieves its global minimum F_r (0.810) and global maximum RMSE (0.384) — the sharpest possible empirical confirmation of C2. Two-sided tails MNAR is substantially less damaging than one-sided mechanisms, a finding with practical implications for study design.

C4 — Systematic F_r vs RMSE comparison. We compute both F_r and RMSE for all four methods (Mean, KNN, MICE, MIGA) across all five datasets, validated over 10 independent seeds with Wilcoxon signed-rank tests. MIGA achieves 1.48–1.97× lower F_r than MICE on Iris and Haberman ($p < 0.001$); MICE achieves 1.4–2.1× lower RMSE than MIGA on every dataset. A key secondary finding: on Glass ($p = 10$), MICE achieves lower F_r than MIGA, revealing that the F_r advantage is not monotone in p but depends jointly on dimensionality and inter-feature correlation.

C5 — Synthetic $p \times \rho$ characterisation. We design a controlled grid experiment on synthetic Gaussian data (Toeplitz covariance, $p \in \{4, 8, 13, 20, 30\}$, $\rho \in \{0, 0.3, 0.6, 0.9\}$) to isolate the effect of dimensionality and correlation on MIGA's F_r advantage. The key finding: MIGA loses when high correlation ($\rho = 0.9$) coincides with moderate-or-higher dimensionality ($p \geq 8$), confirming that MICE's iterative regression becomes an effective joint distribution estimator under these conditions.

C6 — IPW-F_r for MNAR bias correction. We implement Inverse Probability Weighted F_r as an algorithmic correction for MNAR selection bias in X_A. A logistic regression propensity model re-weights X_A rows to correct the biased reference distribution. Experiments across four datasets reveal a critical n/p threshold: when $n/p \gtrsim 20$ with directional MNAR, IPW-F_r reduces F_r by 1–38%; when $n/p \lesssim 14$, the propensity model overfits and increases F_r by up to 35%. The F_r–RMSE orthogonality established in C2 remains intact under IPW.

C7 — Adaptive Diversity Injection (AD-MIGA). The fixed 90% diversity injection rate in MIGA wastes budget late in runs when the elite pool has stabilised. AD-MIGA replaces it with an exponential decay schedule $\alpha(g) = \exp(-\lambda g/G)$, $\lambda = 2.0$, starting at full injection and shifting toward exploitation as the population matures. AD-MIGA-E achieves lower F_r than standard MIGA on all 3 tested datasets (Iris, Haberman, Wholesale) and converges approximately 33 generations earlier on average (105 generations on Iris), while the F_r–RMSE orthogonality remains intact.

C8 — Universal F_r–RMSE orthogonality via GAIN comparison. The formal theorem (C2) is extended beyond MICE to neural methods. GAIN (Generative Adversarial Imputation Networks) mixes a reconstruction loss (pushing toward conditional means) with an adversarial objective (pushing toward distributional fidelity). We prove (Corollary 2.3.2) that no finite $\alpha > 0$ allows GAIN to jointly achieve $F_r = 0$ and RMSE $= 0$. An empirical α-sweep ($\alpha \in \{0, 1, 10, 100, 1000, 10000\}$, all 5 datasets, 5 seeds each) reveals three distinct failure regimes: monotone-but-off-frontier on Iris and Wine, training instability at high α on Haberman, and domination by MICE on both metrics on Glass. The impossibility holds universally across all reconstruction weights and all datasets, elevating C2 to a *universal impossibility theorem* for imputation.

C9 — Parallel execution of independent GA runs. The Q independent runs of MIGA share no state and are trivially parallelisable. We implement `n_jobs` via `concurrent.futures.ProcessPoolExecutor` with `numpy.random.SeedSequence`-spawned seeds, preserving exact reproducibility regardless of parallelism level. Benchmarks across all 5 datasets confirm near-linear scaling: 1.94–1.98× speedup at Q=2. At Q=6, per-seed wall-clock time falls from ∼200 seconds to ∼35 seconds with no change to algorithm behaviour.

1.5 Thesis Organisation

The remainder of this thesis is organised as follows. Chapter 2 reviews missing data mechanisms, classical and modern imputation methods, genetic algorithms, and formally proves the F_r–RMSE orthogonality (Section 2.3), including its extension to neural methods via Corollary 2.3.2. Chapter 3 describes the full MIGA implementation, documents every ambiguous design decision, and presents the IPW-F_r extension (Section 3.5.3), the AD-MIGA adaptive diversity schedule (Section 3.8), and the synthetic dimension methodology (Section 3.6). Chapter 4 reports all experimental results: replication, MNAR evaluation, IPW-F_r comparison, baseline comparison with significance tests, variance preservation, downstream evaluation, the synthetic $p \times \rho$ experiment, AD-MIGA (Section 4.11), and the GAIN comparison with α-sweep (Sections 4.12–4.12.3). Chapter 5 discusses the orthogonality findings, scope conditions, limitations, and all implemented future work.

Chapter 2

Background

2.1 Missing Data Mechanisms

The behaviour of any imputation method depends critically on the mechanism that produces the missing values. Rubin (1976) established the canonical taxonomy of three missing data mechanisms, which determines the conditions under which different imputation strategies are valid [11].

Missing Completely At Random (MCAR). A value is MCAR if the probability of its being missing is independent of both the observed data and the missing data. Formally, if R is the missingness indicator matrix ($R_{ij} = 1$ if X_{ij} is observed, 0 if missing), then MCAR requires $P(R \mid X) = P(R)$. Under MCAR, complete-case analysis is unbiased, though it discards information. MCAR is the most restrictive assumption and rarely holds in practice.

Missing At Random (MAR). A value is MAR if the probability of its being missing may depend on the observed data but not on the missing data itself: $P(R \mid X) = P(R \mid X_{\text{obs}})$. Under MAR, the mechanism is ignorable [11] — valid imputation can be performed using only the observed data, without modelling the missingness mechanism. Most modern imputation methods, including MICE and MIGA, assume MAR.

Missing Not At Random (MNAR). A value is MNAR if its probability of being missing depends on the value itself, even after conditioning on observed data: $P(R \mid X) \neq P(R \mid X_{\text{obs}})$. MNAR is the most challenging case. Examples include income values that are more likely to be missing for high earners, or medical test results missing for patients who did not attend follow-up appointments. Under MNAR, the missingness mechanism is non-ignorable: any imputation method that ignores the mechanism will produce biased results. Chapter 4 provides the first empirical characterisation of MIGA's failure mode under MNAR.

The three mechanisms form a hierarchy of restrictiveness. MCAR implies MAR, and MAR implies the mechanism is ignorable. Van Buuren (2018, §1.2) provides an extensive treatment of how to reason about mechanisms in practice [1].

2.2 Classical Imputation Methods

2.2.1 Complete-Case Analysis

Complete-case analysis (listwise deletion) discards any observation with at least one missing value, performing the analysis on the complete cases only. Under MCAR, this produces unbiased estimates but reduces statistical power. Under MAR or MNAR, it introduces selection bias: the complete cases are a non-representative subset of the population. With 30% missing values spread across features, the fraction of complete rows can fall below 10% even for moderate p. Complete-case analysis is therefore rarely appropriate for real-world datasets.

2.2.2 Single Imputation

Single imputation replaces each missing value with a single plausible estimate. The simplest method substitutes the column mean (for continuous features) or the column mode (for categorical features). Mean imputation is computationally trivial and introduces no bias in the mean itself, but it severely distorts the variance structure: replacing 30% of a column with its mean compresses the variance by approximately $(1 - 0.3)^2 \approx 0.49$, nearly halving it.

KNN imputation [14] replaces each missing value with a weighted average of the k nearest complete observations in the feature space. KNN preserves local structure better than mean imputation and is non-parametric, but its quality depends on the choice of k and the distance metric, and it does not account for the uncertainty introduced by imputation.

Both mean and KNN imputation are single-imputation methods: they produce one complete dataset and treat imputed values as if they were observed. This understates the uncertainty due to missingness and produces overconfident inferences [12].

2.2.3 Multiple Imputation by Chained Equations (MICE)

Multiple imputation generates $M > 1$ complete datasets by drawing imputed values from the predictive distribution $P(X_{\text{mis}} \mid X_{\text{obs}})$, combines the analysis results across datasets using Rubin's rules, and produces inferences that properly account for missingness uncertainty [12].

MICE [2], also known as Fully Conditional Specification (FCS), is the dominant multiple imputation method. For a dataset with p partially observed variables, MICE iterates a cycle of p conditional imputation models: for each variable j, it fits a regression model of X_j on all other variables X_{-j} using the currently completed data, then draws imputations from the posterior predictive distribution of this model. The cycle repeats until convergence.

The key strength of MICE is its flexibility: each variable can use a different model family (linear regression for continuous variables, logistic regression for binary variables). MICE is theoretically valid under MAR and produces imputations with the correct conditional distribution of each variable given the others.

However, MICE's optimality criterion is minimum expected squared error, which corresponds to imputing conditional means. Van Buuren (2018, §2.6) explicitly states:

> "The minimum mean squared error is achieved by regression imputation, which suppresses variance and produces biased statistical inference."

This is a structural property: by imputing $\mathbb{E}[X_j \mid X_{-j}]$, MICE systematically underestimates the marginal variance $\text{Var}(X_j)$. By the law of total variance:

$$\text{Var}(X_j) = \mathbb{E}[\text{Var}(X_j \mid X_{-j})] + \text{Var}(\mathbb{E}[X_j \mid X_{-j}]) \tag{2.1}$$

MICE discards the first term. For analyses that require the correct marginal distribution, MICE produces biased results. This is the fundamental motivation for MIGA.

2.3 The F_r–RMSE Orthogonality

A central claim of this thesis is that distributional fidelity (measured by F_r) and pointwise accuracy (measured by RMSE) are *orthogonal* objectives: optimising one systematically degrades the other. This section formalises that claim, first by establishing a general mathematical framework, then by proving the orthogonality as a consequence.

2.3.1 Mathematical Framework

Setup. Let $X \in \mathbb{R}^{n\times p}$ be a dataset with n observations and p features. Let $M \in \{0,1\}^{n\times p}$ be the missingness mask ($M_{ij} = 0$ if X_{ij} is missing). Under MAR, $M \perp X_{\text{mis}} \mid X_{\text{obs}}$. An *imputation map* is any measurable function $\mathcal{F} : (X_{\text{obs}}, M) \mapsto \hat{X}_{\text{mis}}$.

Definition 2.1 (Irreducible conditional variance)**.** For feature j with at least one missing value, define the *irreducible conditional variance*:

$$\tau_j \;=\; \mathbb{E}\big[\mathrm{Var}(X_j \mid X_{-j})\big] \;=\; \mathrm{Var}(X_j)\,(1 - R_j^2), \tag{2.2}$$

where R_j^2 is the population coefficient of determination of X_j regressed on X_{-j}. Write $\tau = \sum_{j\in\mathcal{J}} \tau_j$ where $\mathcal{J}$ is the set of features with missing values.

τ_j is the variance of X_j that cannot be recovered from the remaining features: it is the *aleatoric* uncertainty of imputation. When $R_j^2 \approx 1$ (feature j is nearly linearly predictable from X_{-j}), $\tau_j \approx 0$ and RMSE-optimal imputation is nearly distributional-faithful; when $R_j^2 \approx 0$ (feature j is independent of X_{-j}), $\tau_j = \mathrm{Var}(X_j)$ and the deficiency is maximal.

RMSE decomposition. For any imputation map $\mathcal{F}$, the bias–variance decomposition of squared error gives:

$$\mathbb{E}\big[\mathrm{RMSE}(\mathcal{F})^2\big] = \underbrace{\mathbb{E}\big[\|\mathcal{F} - \mathbb{E}[X_{\text{mis}} \mid X_{\text{obs}}]\|^2\big]}_{\text{estimation error} \geq 0} + \underbrace{\tau}_{\text{irreducible variance}}. \tag{2.3}$$

The estimation error is non-negative and vanishes only when $\mathcal{F} = \mathbb{E}[X_{\text{mis}} \mid X_{\text{obs}}]$. Therefore the unique RMSE-minimising imputer is:

$$\mathcal{F}^* \;=\; \mathbb{E}[X_{\text{mis}} \mid X_{\text{obs}}], \qquad \mathrm{RMSE}^* \;=\; \sqrt{\tau}, \tag{2.4}$$

and no imputer can achieve RMSE $< \sqrt{\tau}$.

Theorem 2.1 (Quantitative F_r lower bound for MMSE imputation). *Let $\pi_j \in (0, 1]$ be the proportion of rows in X_C for which feature j is missing, and let R_j^2 be as in Definition 2.1. Under the MMSE imputer $\mathcal{F}^*$:*

$$F_r(\mathcal{F}^*) \;\geq\; D_r\big(\tilde{S}(\mathcal{F}^*),\, I\big) \;\geq\; \max_{j \in \mathcal{J}} \; \pi_j \, (1 - R_j^2). \tag{2.5}$$

Proof. For $j \in \mathcal{J}$, the completed block X_C under $\mathcal{F}^*$ mixes two sub-populations: the $(1 - \pi_j)$ fraction of rows in which X_j is observed (contributing marginal variance $\mathrm{Var}(X_j)$) and the π_j fraction in which X_j is imputed via $\mathbb{E}[X_j \mid X_{-j}]$ (contributing variance $R_j^2 \, \mathrm{Var}(X_j)$). Under MAR the two sub-populations share the same mean, so the sample variance of column j after MMSE imputation satisfies:

$$[S_C(\mathcal{F}^*)]_{jj} \;\approx\; [S_A]_{jj} \, \big(1 - \pi_j(1 - R_j^2)\big).$$

The relative covariance $\tilde{S} = S_A^{-1/2} S_C S_A^{-1/2}$ therefore has diagonal entry $\tilde{S}_{jj} \approx 1 - \pi_j(1 - R_j^2)$, so $|\tilde{S}_{jj} - 1| = \pi_j(1 - R_j^2)$. Since $D_r(\tilde{S}, I) \geq |\tilde{S}_{jj} - 1|$ for any j, and $F_r \geq D_r(\tilde{S}, I)$, the bound follows. $\square$

Remark. Theorem 2.1 has a direct empirical interpretation. The bound $\pi_j(1 - R_j^2)$ is largest when a feature has both high missingness and low predictability from other features. On Haberman, the Survival feature has $R_j^2 \approx 0.07$ and $\pi_j = 0.30$, giving a bound of ≈ 0.28; this explains the large empirical F_r gap between MIGA and MICE in Chapter 4. On Glass, strong inter-feature correlations ($R_j^2 \to 1$ for most features) suppress the bound, explaining why MICE achieves competitive F_r despite its MMSE objective — the variance deficiency is small because the features are nearly deterministically predictable from one another.

Proposition 2.1 (RMSE cost of proper imputation). *Let $\hat{X}_{\text{mis}} \sim P(X_{\text{mis}} \mid X_{\text{obs}})$ be a* proper *imputation, drawn independently of the true X_{mis} (van Buuren, 2018, §3.4). Then:*

$$\mathbb{E}\big[\text{RMSE}^2\big] \;=\; 2\tau. \tag{2.6}$$

That is, proper imputation achieves exactly $\sqrt{2}$ times the minimum RMSE.

Proof. Given X_{obs}, both X_{mis} and $\hat{X}_{\text{mis}}$ are independent draws from $P(X_{\text{mis}} \mid X_{\text{obs}})$. Therefore:

$$\mathbb{E}\big[\|X_{\text{mis}} - \hat{X}_{\text{mis}}\|^2 \mid X_{\text{obs}}\big] = \text{Var}(X_{\text{mis}} \mid X_{\text{obs}}) + \text{Var}(\hat{X}_{\text{mis}} \mid X_{\text{obs}}) = 2\,\text{Var}(X_{\text{mis}} \mid X_{\text{obs}}),$$

where the cross-term vanishes by $X_{\text{mis}} \perp \hat{X}_{\text{mis}} \mid X_{\text{obs}}$. Taking the outer expectation gives $\mathbb{E}[\text{RMSE}^2] = 2\tau$. □

Theorem 2.2 (F_r–RMSE trade-off bound). *Let $\tau > 0$. Then:*

(i) The minimum achievable RMSE is $\sqrt{\tau}$, attained uniquely by $\mathcal{F}^*$.

(ii) At $\mathcal{F}^*$: $F_r \geq \max_{j \in \mathcal{J}} \pi_j (1 - R_j^2) > 0$.

(iii) Any $\mathcal{F}$ achieving $F_r = 0$ must have $\text{RMSE} > \sqrt{\tau}$.

(iv) The point $(F_r, \text{RMSE}) = (0, \sqrt{\tau})$ is unachievable.

Proof. Parts (i) and (ii) follow from Eq. (2.4) and Theorem 2.1. For (iii): from part (ii), $F_r(\mathcal{F}^*) > 0$, so any $\mathcal{F}$ with $F_r(\mathcal{F}) = 0$ must differ from $\mathcal{F}^*$. By the RMSE decomposition

(2.3), any $\mathcal{F} \neq \mathcal{F}^*$ has $\mathrm{RMSE}(\mathcal{F})^2 = \|\mathcal{F} - \mathcal{F}^*\|^2 + \tau > \tau$, so $\mathrm{RMSE}(\mathcal{F}) > \sqrt{\tau}$. Part (iv) follows from (i) and (iii). □

Remark. Proposition 2.1 shows that proper imputation — the ideal distributional-fidelity imputer — pays a factor of $\sqrt{2}$ in RMSE relative to the minimum. MIGA approximates proper imputation via the GA: its RMSE penalty over MICE (1.4–2.1× empirically; Chapter 4) is consistent with this $\sqrt{2}$ theoretical bound, with the excess arising from imperfect distributional matching.

Theorems 2.1 and 2.2 together constitute the *quantitative* form of the orthogonality. The propositions below restate the qualitative existence claims that follow as immediate corollaries.

2.3.2 Orthogonality Propositions

Proposition 2.2 (RMSE-optimal imputation has positive F_r)**.** *Let X_C be the block of rows with at least one missing value, and let $\hat{X}_C^* = \arg\min_{\hat{X}} \mathbb{E}[\mathrm{RMSE}(\hat{X}, X_C)]$ be the RMSE-minimising imputation. If the conditional variance $\mathrm{Var}(X_j \mid X_{-j}) > 0$ for any feature j with missing values, then $F_r(\hat{X}_C^*) > 0$.*

Proof. The RMSE-minimising imputation sets each missing value to its conditional expectation:

$$\hat{X}_{ij}^* = \mathbb{E}[X_{ij} \mid X_{i,\mathrm{obs}}].$$

By the law of total variance, for any feature j:

$$\mathrm{Var}(X_j) = \underbrace{\mathbb{E}[\mathrm{Var}(X_j \mid X_{-j})]}_{\text{within-group variance}} + \underbrace{\mathrm{Var}(\mathbb{E}[X_j \mid X_{-j}])}_{\text{between-group variance}} .$$

The conditional-expectation imputation retains only the between-group term. Since the within-group variance is strictly positive by assumption, the imputed column has strictly lower marginal variance than the true column:

$$\mathrm{Var}(\hat{X}_j^*) = \mathrm{Var}(\mathbb{E}[X_j \mid X_{-j}]) < \mathrm{Var}(X_j) \approx \mathrm{Var}(X_j^A),$$

where X_j^A denotes column j of the available rows X_A. The relative covariance matrix $\tilde{S} = S_A^{-1/2} S_C S_A^{-1/2}$ therefore deviates from the identity, so $D_r(\tilde{S}, I) > 0$, and since $F_r \geq D_r(\tilde{S}, I)$ it follows that $F_r(\hat{X}_C^*) > 0$. □

Proposition 2.3 (F_r-zero imputation has positive RMSE)**.** *Let $\hat{X}_C^{**}$ be any imputation satisfying $F_r(\hat{X}_C^{**}) = 0$. Then $\mathbb{E}[\mathrm{RMSE}(\hat{X}_C^{**}, X_C)] > 0$ unless the true missing values are drawn from the same distribution as X_A.*

Proof. $F_r = 0$ requires $\hat{X}_C^{**}$ to match X_A in mean, covariance, and skewness — a constraint on the *marginal distribution* of the imputed rows. RMSE, by contrast, depends on the *joint distribution* of ($\hat{X}_{ij}^{**}$, X_{ij}): specifically, RMSE $= 0$ requires $\hat{X}_{ij}^{**} = X_{ij}$ for every missing cell. A method can achieve $F_r = 0$ by sampling from X_A's empirical distribution independently of the true missing values, producing RMSE > 0 in expectation. Equality RMSE $= 0$ additionally requires each imputed value to equal the corresponding true value, a strictly stronger condition than marginal distributional matching. Therefore $F_r = 0$ and RMSE $= 0$ cannot simultaneously hold unless the true X_C is drawn from exactly the same distribution as X_A. □

Corollary 2.3.1 (Objectives are non-collinear)**.** *No imputation method can simultaneously minimise $\mathbb{E}[\mathrm{RMSE}]$ and F_r in general. Methods that minimise RMSE (e.g. MICE) will generically have $F_r > 0$; methods that minimise F_r (e.g. MIGA) will generically have RMSE above the* *irreducible minimum.*

Remark. The orthogonality does not imply that F_r and RMSE are *uncorrelated* across methods. In practice, both tend to decrease as an imputation method improves. The claim is that *optimising one does not optimise the other*: there is no single imputation that jointly achieves the minimum of each objective. This is the formal basis for the empirical scatter observed in Chapter 4, Figure 4.1.

The practical consequence is that the choice between MICE and MIGA is not a question of which method is universally better, but which *objective* is appropriate for the downstream application. When the analysis requires the correct marginal distribution — density estimation, distributional hypothesis tests, or variance-sensitive statistics — F_r is the appropriate criterion and MIGA is preferred. When the goal is accurate reconstruction of individual missing values — prediction tasks where imputed cells are directly evaluated — RMSE is the appropriate criterion and MICE or KNN may be preferred. Chapter 4 provides empirical evidence that MIGA's advantage over MICE on F_r is consistent and statistically significant under MAR (Section 4.6), while MICE retains a RMSE advantage, confirming Corollary 2.3.1 in practice.

2.3.3 Extension to Neural Imputation Methods

Propositions 2.2 and 2.3 formally apply to any method that either (a) minimises RMSE or (b) achieves $F_r = 0$. This subsection examines how GAIN [15] — a GAN-based neural imputer — relates to this theory.

GAIN's objective. GAIN trains a generator G to produce imputed values that fool a discriminator D, with an additional MSE reconstruction term on observed positions:

$$\mathcal{L}_G = \underbrace{\mathbb{E}\left[(1-M)\cdot\log D(\hat{X},H)\right]}_{\text{adversarial}} + \alpha\,\underbrace{\mathbb{E}\left[M\cdot\|X-G(\tilde{X},M)\|^2\right]}_{\text{MSE on observed}} \tag{2.7}$$

where $\alpha > 0$ is the reconstruction weight, M is the observed mask, and H is a hint matrix.

Interpolation interpretation. GAIN with $\alpha \to \infty$ forces G to replicate observed values exactly; the discriminator provides no additional signal. In this regime, GAIN effectively reduces to a conditional mean imputer (MICE-like): low RMSE but $F_r > 0$ by Proposition 2.2. GAIN with $\alpha \to 0$ removes the MSE constraint; the generator is guided purely by the adversarial term, which pushes toward correct distributional matching. In this regime, GAIN approaches a proper posterior sampler: lower F_r but higher RMSE.

Any finite $\alpha > 0$ therefore places GAIN on the Fr–RMSE Pareto frontier between these extremes. The key claim is:

Corollary 2.3.2 (GAIN cannot achieve $F_r = 0$ and RMSE $= 0$ simultaneously)**.** *For any $\alpha >$ 0, GAIN's MSE component biases the generator toward conditional means, so $F_r > 0$ by the same law-of-total-variance argument as Proposition 2.2. Simultaneously, the adversarial component prevents G from being a pure conditional-mean estimator, so RMSE > 0. GAIN therefore cannot jointly achieve $F_r = 0$ and* RMSE $= 0$.

Remark. Corollary 2.3.2 does *not* claim that GAIN lies on the Fr–RMSE Pareto frontier between MIGA and MICE. At $\alpha = 100$ (the paper's default), experiments (Section 4.12) show GAIN is *dominated* by MICE on Iris: higher F_r (1.478 vs 1.271) *and* higher RMSE (0.124 vs 0.101). The adversarial component adds variance to the imputed values without directing them toward the correct distribution, resulting in off-frontier performance.

The corollary's claim is solely the impossibility: no $\alpha > 0$ allows GAIN to achieve both objectives simultaneously.

This extends the impossibility from the MIGA/MICE pair to any method mixing reconstruction and adversarial objectives. Section 4.12 provides empirical confirmation.

2.4 Genetic Algorithms

2.4.1 Foundations

A Genetic Algorithm (GA) is a population-based metaheuristic search method inspired by Darwinian natural selection [5]. A GA maintains a population of candidate solutions, evaluates each candidate using a fitness function, and iteratively produces better solutions by applying selection, crossover (recombination), and mutation operators.

The canonical GA operates as follows [4]:

1. **Initialisation:** Generate an initial population of l individuals, typically at random.
2. **Evaluation:** Compute the fitness $f(x)$ for each individual x.
3. **Selection:** Select individuals for reproduction with probability proportional to fitness.
4. **Crossover:** With probability p_c, combine pairs of selected individuals to produce offspring.
5. **Mutation:** With probability p_m, randomly alter components of offspring.
6. **Replacement:** Form the new population from survivors and offspring.
7. **Repeat** steps 2–6 for G generations or until convergence.

GAs are particularly suited to problems where the fitness landscape is discontinuous, multimodal, or not differentiable — properties that make gradient-based methods inapplicable.

2.4.2 MIGA's Genetic Algorithm

MIGA [3] uses a custom GA that departs from the canonical scheme. Most notably, MIGA uses an aggressive diversity injection: at each generation, the majority of the population (approximately 90% under the paper's default parameters) is replaced with randomly generated individuals drawn from per-variable generators R_j. This effectively restarts the search from random each generation, preserving only the best c individuals.

The GA optimises the fitness function F_r over the space of possible imputed values for the missing cells. Each individual in the population represents one complete candidate imputation: a vector of values for all missing cells in the dataset. The fitness function F_r measures distributional distance between the available rows X_A and the completed rows X_C, penalising divergence in means, covariance structure, and skewness.

2.4.3 Exploration-Exploitation in MIGA

The paper parameterises the operator outputs as:

- c: elite individuals carried forward unchanged
- $c_1 \times c_3$: mutated copies of elite individuals
- c_2: crossover offspring from pairs of elites
- $l - c - c_1c_3 - 2(c_2 - 1)$: random individuals from generators (diversity injection)

Under the paper's default settings ($l = 200$, $c = 3$, $c_1 = 3$, $c_2 = 2$, $c_3 = 5$), the diversity injection fills 180/200 = 90% of the population. This unusual balance means that MIGA is closer to a repeated random restart with elite preservation than a conventional GA.

2.5 The MIGA Paper

Figueroa-García, Neruda, and Hernandez-Pérez (2023) proposed MIGA in "A genetic algorithm for multivariate missing data imputation," published in *Information Sciences* (619, 947–967) [3].

2.5.1 Problem Formulation

Given a dataset X with n rows and p features, MIGA partitions it into X_A (the n_A rows with no missing values) and X_C (the $n_C = n - n_A$ rows with at least one missing value). The goal is to find imputed values for the missing cells of X_C such that the statistical properties of the completed X_C match those of X_A.

2.5.2 The Fitness Function F_r

The fitness function F_r quantifies the distributional distance between X_A and X_C using three terms:

$$F_r = D_r(\tilde{x}_A, \tilde{x}_C) + D_r(\tilde{S}, I) + D_r(b_A, b_C) \tag{2.8}$$

where D_r denotes the Minkowski distance of order r (typically $r = \infty$, the Chebyshev norm), and:

- $\tilde{x} = \bar{x}/S_p$ is the standardised mean vector

- $\tilde{S} = S_A^{-1/2} S_C S_A^{-1/2}$ is the relative covariance matrix (equal to the identity when $S_C = S_A$)
- b is the bias-corrected skewness vector

$F_r = 0$ if and only if X_C has the same mean, covariance, and skewness as X_A.

2.5.3 Per-Variable Generators

Each variable j has a random generator R_j used to sample candidate values. The paper specifies that generators are "obtained from samples" of the observed values of variable j, which this thesis interprets as bootstrap resampling from the empirical distribution (§3.2.5). For integer-valued or discrete variables, this ensures that imputed values are always valid integers.

2.5.4 Gaps in the Original Paper

Three gaps motivate this thesis:

1. **No public implementation.** The paper contains no source code, and implementation details are underspecified at several points.
2. **Weak baseline comparison.** CMIM (2005) and ANNI are significantly weaker than modern methods. No comparison against KNN or MICE is provided.
3. **No MNAR evaluation.** All experiments assume MAR. The behaviour of MIGA's fitness function under non-ignorable missingness is unknown.

2.6 Related Work

2.6.1 Distributional Imputation

The observation that RMSE-optimal imputation suppresses variance has motivated several lines of work. Siddique and Belin (2008) proposed predictive mean matching as a way to preserve distributional properties by imputing observed values rather than model predictions [13]. Van Buuren (2018, §3.4) discusses "proper imputation" — drawing from the full posterior predictive distribution rather than its mean — as a way to avoid variance suppression [1]. MIGA can be understood as a non-parametric alternative to proper imputation.

Muzellec et al. (2020) proposed optimal transport-based imputation, which finds imputed values that minimise the Sinkhorn divergence between the completed and observed distributions [9] — directly analogous to MIGA's F_r objective, but using a differentiable surrogate. They note that distributional methods will generically achieve higher RMSE than regression-based methods, consistent with the findings of this thesis.

2.6.2 Neural Network Imputation

GAIN [15] uses a Generative Adversarial Network to generate imputations, with a discriminator that learns to distinguish imputed from observed values — conceptually similar to MIGA's distributional objective but in a learned feature space. GRAPE [16] formulates imputation as a graph completion problem using message passing over a bipartite graph of observations and features. A comparison against GAIN or GRAPE is identified as future work (§5.6.3).

2.6.3 Evaluation of Imputation Methods

Oberman and Vink (2024) survey evaluation practices for imputation and find no universal standard for RMSE normalisation, making cross-paper comparisons unreliable [10]. This thesis uses range-normalised RMSE to match the paper's reported baseline values. Ipsen et al. (2022) show that evaluating imputation quality under MNAR using RMSE on the missing cells is itself biased, because the missing cells are a non-representative sample of the marginal distribution [7] — a point directly relevant to Chapter 4's MNAR analysis.

Chapter 3

Methodology

3.1 Overview

This chapter describes the complete implementation of MIGA from the paper specification, documents every decision made where the paper is ambiguous or silent, and presents the novel extensions introduced in this thesis: IPW-F_r for MNAR bias correction (C6, Section 3.5.3) and the synthetic dimension experiment methodology (C5, Section 3.6). The formal F_r–RMSE orthogonality result (Chapter 2, Propositions 2.2–2.3) provides the theoretical framework within which all contributions are situated.

All code is implemented in Python 3.12 using NumPy, SciPy, and scikit-learn. The implementation is structured as a Python package (`miga/`) with the modules described in Table 3.1.

TABLE 3.1: Python package module structure.

Module	Responsibility
`miga/statistics.py`	Statistical primitives: `compute_stats`, `compute_kurtosis`, `ledoit_wolf_cov`
`miga/fitness.py`	`FitnessEvaluator`: pre-computes X_A statistics, evaluates F_r (with optional IPW weighting)
`miga/operators.py`	`initialize_population`, `mutate`, `crossover`, `diversity`
`miga/core.py`	`MIGA`: top-level estimator, orchestrates Q runs
`miga/data_utils.py`	Dataset loaders, `apply_mar`, `apply_mnar`, `auto_generators`

3.2 Reimplementation of MIGA (C1)

3.2.1 Population Representation

Each individual in the genetic algorithm population represents a complete candidate imputation for all missing cells in the dataset. If the dataset has k missing cells (across all rows in X_C), an individual is a vector of k real numbers. The mapping between positions in this vector and (row, column) locations in the dataset is stored as a list `missing_index` of (local_row, col) pairs, constructed once before the GA begins.

This flat vector representation is efficient: fitness evaluation requires only reassembling X_C from the current individual and computing the three statistical distances. The population is stored as a 2D array of shape (l, k).

3.2.2 Fitness Evaluation

The fitness function F_r is evaluated by `FitnessEvaluator`, which pre-computes all X_A statistics at initialisation (mean, covariance, skewness, and optionally kurtosis) and evaluates each candidate X_C efficiently. Feature scaling is applied before all computations (§3.2.3).

The three terms follow Equations 5–10 of Figueroa-García et al. (2023) [3]:

d_{means} — Standardised mean distance:

$$d_{\text{means}} = D_r(\tilde{x}_A, \tilde{x}_C)\,, \quad \tilde{x} = \frac{\bar{x}}{S_p} \tag{3.1}$$

where S_p is the pooled standard deviation vector:

$$S_{p,j}^2 = \frac{\mathrm{diag}(S_A)_j \cdot \nu_A + \mathrm{diag}(S_C)_j \cdot \nu_C}{\nu_A + \nu_C} \tag{3.2}$$

with $\nu_A = n_A - 1, \nu_C = n_C - 1$.

$d_{\mathbf{cov}}$ — Relative covariance distance:

$$d_{\mathrm{cov}} = D_r\big(\tilde{S},\, I\big)\,, \quad \tilde{S} = S_A^{-1/2} S_C S_A^{-1/2} \tag{3.3}$$

$\tilde{S} = I$ if and only if $S_C = S_A$. The matrix square root is computed via eigendecomposition: $S_A^{-1/2} = V \Lambda^{-1/2} V^\top$.

$d_{\mathbf{skew}}$ — Skewness distance:

$$d_{\mathrm{skew}} = D_r(b_A,\, b_C) \tag{3.4}$$

where b_j is the bias-corrected sample skewness of column j.

All distances use the Minkowski order r specified in the paper (typically $r = \infty$, the Chebyshev norm: $D_\infty(A, B) = \max_{i,j} |a_{ij} - b_{ij}|$).

3.2.3 Feature Scaling

Problem. Wine's Proline feature has variance $\sigma^2 \approx 10^5$ while other features have $\sigma^2 \approx 1$. Without scaling, the condition number of S_A is approximately 4×10^7, making $S_A^{-1/2}$ numerically meaningless and $d_{\mathrm{cov}}(X_A, X_A) \neq 0$.

Solution. Before computing any statistics, divide each column of X_A and X_C by $\hat{\sigma}_j = \mathrm{std}(X_A[:, j])$ (estimated from X_A alone, to avoid data leakage from X_C). This transforms S_A into a correlation matrix with eigenvalues bounded by $O(p)$.

After scaling, Wine 30% condition number drops from 4×10^7 to 158 and $d_{\text{cov}}(X_A, X_A) = 0.000$.

3.2.4 Eigenvalue Floor in Relative Covariance

Even after feature scaling, S_A can be rank-deficient when $n_A < p$. In `relative_cov()`, eigenvalues of S_A are floored at $\max(\lambda_{\max} \times 10^{-4}, 10^{-10})$ before computing $\Lambda^{-1/2}$. This bounds the condition number of $S_A^{-1/2}$ at 10^2, preventing catastrophic noise amplification.

3.2.5 Bootstrap Generators

The paper specifies that generators R_j are "obtained from samples" of the observed values of variable j. We interpret this as bootstrap resampling from the empirical distribution: R_j draws uniformly from the set of non-missing observed values of column j.

For integer-valued columns, the generator returns integer samples only. This is essential for datasets with discrete features: Haberman's Nodes column has 44% zeros, and a Gaussian generator would never generate zero. The bootstrap generator samples zeros with probability 0.44, correctly preserving the discrete marginal distribution.

3.2.6 Genetic Algorithm Operators

Selection. The top-c individuals (lowest F_r) are retained as elites unchanged.

Mutation. Each of the c_1 elites produces c_3 mutated copies. A mutated copy is identical to its parent except that one randomly chosen component (one missing cell value) is replaced by a fresh sample from R_j for the corresponding column j.

Crossover. c_2 crossover offspring are produced by uniform crossover of random pairs drawn from the elites. Each component of the offspring is taken from one parent with probability 0.5.

Diversity injection. The remaining $l - c - c_1c_3 - 2(c_2 - 1)$ population slots are filled with fresh random individuals sampled entirely from the generators. Under the paper's default parameters ($l = 200$, $c = 3$, $c_1 = 3$, $c_2 = 2$, $c_3 = 5$), this is 180 individuals, or 90% of the population.

3.2.7 Multi-Run Strategy

MIGA runs Q independent GA instances on the same missing dataset, each with a different internal random seed. The best individual across all Q runs is returned as the final imputation. `miga.generation_history_` stores the full convergence curve (best F_r per generation) for each run, enabling convergence analysis.

3.3 Ledoit-Wolf Shrinkage Covariance (C2)

3.3.1 The Rank-Deficiency Problem

The relative covariance term $d_{\text{cov}} = D_r(S_A^{-1/2} S_C S_A^{-1/2}, I)$ requires computing $S_A^{-1/2}$, which is undefined when S_A is rank-deficient. S_A becomes rank-deficient when $n_A < p$.

For Wine at 40% missing: $n_A = 11$, $p = 13$. The sample covariance has condition number 5.7×10^8 and $d_{\text{cov}}(X_A, X_A) = 0.606$, a mathematical impossibility since d_{cov} should equal zero when both arguments are the same set.

3.3.2 The Ledoit-Wolf Estimator

The Ledoit-Wolf (2004) shrinkage estimator [8] replaces the sample covariance with a convex combination:

$$S_{\text{LW}} = (1-\alpha)\, S_{\text{sample}} + \alpha \cdot \frac{\text{tr}(S_{\text{sample}})}{p} \cdot I \tag{3.5}$$

where $\alpha \in [0, 1]$ is the shrinkage coefficient chosen analytically. The target $(\text{tr}(S)/p) \cdot I$ is always positive definite, so S_{LW} is positive definite for any $\alpha > 0$.

We apply the Ledoit-Wolf estimator to **both** S_A and S_C. Applying it to S_A alone would break the invariant $d_{\text{cov}}(X_A, X_A) = 0$.

3.3.3 Results

TABLE 3.2: Wine: sample vs Ledoit-Wolf covariance at increasing missing rates.

Missing	n_A	**Sample cond.**	**LW cond.**	d_{cov} **sample**	d_{cov} **LW**
30%	23	158	9.1	0.000 ✓	0.000 ✓
40%	11	570,000,000	8.8	0.606 ×	0.000 ✓

Ledoit-Wolf restores mathematical consistency at 40% missing. RMSE is not improved — $n_A = 11$ is an insufficient sample size to reliably estimate the joint distribution of 13 features, regardless of the covariance estimator.

3.4 Budget-Aware Restarts with Adaptive Diversity Decay (C3)

3.4.1 Motivation

Standard MIGA allocates a fixed budget of Q runs each of G generations. However, fitness curves in practice converge within 100–300 generations on most datasets, leaving the remainder of each run idle. On Haberman, for example, the GA reaches its best F_r within the first $\sim$150 generations and then stagnates, wasting 70% of the budget.

The diversity injection mechanism fills 90% of the population with random individuals at every generation. This is well-suited for exploration early in a run, but is counter-productive late in a run once the elite solutions have converged: fresh random individuals can only dislodge the elite by chance. At that point, more restarts with fresh populations would be more productive than continued random injection around a stagnated solution.

3.4.2 Implementation

Two components work together:

Early stopping. Each run tracks a stagnation counter incremented whenever the best F_r fails to improve by more than $\varepsilon = 10^{-8}$. When the counter reaches the patience threshold P, the run terminates:

$$\text{stop run if } \min_{g' \leq g} F_r(g') - F_r(g) < \varepsilon \text{ for } P \text{ consecutive generations} \tag{3.6}$$

Budget-aware restarts. The total generation budget $B = Q \times G$ is fixed. Instead of exactly Q runs of length G, the algorithm launches restarts in sequence until the budget is exhausted:

$$\text{while} \sum_{\text{runs}} |\text{gen\_history}| < B : \quad \text{launch new run} \tag{3.7}$$

Each run is capped at $\min(G,\ B - \text{gens\_used})$ generations so the total never exceeds B.

Adaptive diversity decay. Within each run, the number of random (diversity) individuals decays linearly from $n_{\text{div}}^{\max}$ (90% of l) at generation 0 to $(1 - \delta) \cdot n_{\text{div}}^{\max}$ at generation G, with $\delta = 0.7$:

$$n_{\text{div}}(g) = \left\lfloor n_{\text{div}}^{\max} \cdot \left(1 - \delta \cdot \frac{g}{G-1}\right) \right\rfloor \tag{3.8}$$

Slots freed by reduced diversity are filled with additional mutation offspring from the elite pool, progressively shifting the population from exploration to exploitation as each run matures. When early stopping triggers before full decay, the harmful late-stage exploitation phase is avoided.

Parameters: $P = 50$, $\delta = 0.7$, activated via `MIGA(early_stopping_patience=50, diversity_decay=0.7)`.

3.4.3 Why the Components Need Each Other

Diversity decay alone was tested at full $G = 500$ and worsened RMSE by 14–16% on Haberman: late-stage exploitation of a stagnated run amplifies a poor local optimum. Early stopping alone gives more restarts but each run still spends its later generations doing 90% random injection around a converged solution. Together: early stopping prevents the harmful late-decay phase, while decay ensures that the fraction of the budget spent near good solutions grows over the run's lifetime.

3.5 MNAR Evaluation Framework (C4)

3.5.1 Implementation of MNAR Mechanisms

The paper evaluates MIGA only under MAR. We implement three MNAR mechanisms in `apply_mnar(X, pct, mechanism)`:

top: For each selected feature j, the n_{remove} rows with the highest values of $X_{:,j}$ are set missing.

bottom: The n_{remove} rows with the lowest values are set missing.

tails: Half the removals come from each extreme: half $= \max(1, \lfloor n_{\text{remove}}/2 \rfloor)$, removing from both the top and bottom of the distribution.

3.5.2 Why MNAR Breaks MIGA's Objective

Under MAR, X_A is a representative sample of the full population. The fitness function's goal — make X_C look like X_A — is equivalent to making X_C look like the true population.

Under MNAR `top`, rows with high values of the selected features go missing, so X_A contains only rows with low values for those features. The fitness function instructs the GA to match X_C to this biased X_A. The GA succeeds — it achieves a very low F_r — but the imputed values are systematically too low, because the true missing values were the high-value rows.

3.5.3 IPW-F_r Implementation (C6)

The standard correction for selection bias is inverse probability weighting (IPW) [6]. Let $\pi_i = P(R_i = 1 \mid X_i)$ be the propensity score for row i being complete. The IPW fitness function replaces the unweighted X_A statistics with sample-weighted equivalents:

$$F_r^{\mathrm{IPW}} = D_r(\tilde{x}_A^w, \tilde{x}_C) + D_r\big(\tilde{S}^w, I\big) + D_r(b_A^w, b_C) \tag{3.9}$$

where superscript w denotes statistics computed with weights $w_i = 1/\pi_i$, up-weighting under-represented complete rows and down-weighting over-represented ones.

Propensity estimation. Since π_i is unknown, we estimate it by logistic regression: treat the binary completeness indicator $R_i \in \{0, 1\}$ as the response and all features (with column-mean imputation for missing entries) as predictors. Scikit-learn's `LogisticRegression` (`C=1.0`, L-BFGS solver) is fitted on the full dataset before the GA begins. Propensity scores are floored at $\pi_{\min} = 0.05$ to prevent extreme weights, and weights are normalised to sum to n_A so the weighted statistics are on the same scale as the unweighted ones.

Weighted statistics. Three new primitives are added to `miga/statistics.py`:

- `weighted_mean(X, w)`: $\bar{x}_j^w = \sum_i w_i X_{ij} / \sum_i w_i$
- `weighted_cov(X, w)`: via `numpy.cov(aweights=w)`, which applies the reliability-weight correction (denominator $\sum w - \sum w^2 / \sum w$), the weighted analogue of ddof=1.
- `weighted_skewness(X, w)`: using effective sample size $n_{\mathrm{eff}} = (\sum w)^2 / \sum w^2$ for the Fisher bias-correction factor $\sqrt{n_{\mathrm{eff}}(n_{\mathrm{eff}} - 1)}/(n_{\mathrm{eff}} - 2)$, matching

`scipy.stats.skew(bias=False)` at uniform weights.

The `FitnessEvaluator` class accepts an optional `ipw_weights` array; when provided, it replaces `compute_stats(X_A)` with the three weighted primitives. The top-level `MIGA` estimator exposes `use_ipw=True` to activate this path.

Computational cost. Propensity estimation adds one logistic regression fit ($O(n \cdot p)$) before the GA, negligible relative to $G = 500$ generations. Weighted statistics add no asymptotic overhead versus unweighted ones.

Applicability condition. Logistic regression propensity models require $n/p \gtrsim 20$ for stable estimates. When $n/p \lesssim 14$ (e.g. Wine, $n = 178$, $p = 13$), the model overfits the completeness indicator, producing extreme propensity scores that destabilise the weighted covariance and increase F_r. For such datasets, IPW-F_r should be applied with a penalised estimator (ridge logistic regression) or omitted. The correction is most effective when the MNAR mechanism creates a directional selection bias (top- or tail-censoring) and $n/p \gtrsim 20$; for uniform missingness (MAR) or symmetric mechanisms (bottom-censoring), re-weighting adds estimation variance without correcting bias.

3.6 Synthetic Dimension Experiment (C5)

3.6.1 Motivation

The five UCI benchmark datasets provide good empirical coverage, but each dataset has a fixed combination of dimensionality (p), correlation structure (ρ), and distributional shape that cannot be disentangled from each other. To isolate the effect of p and ρ on MIGA's distributional advantage, we design a controlled synthetic experiment.

Section 2.3 predicts that MIGA's F_r advantage should degrade with increasing p: as the number of features grows, the moment-matching objective becomes a coarser approximation of the full joint distribution, and the fitness landscape for the GA becomes harder to navigate. The synthetic experiment tests this prediction directly.

3.6.2 Data Generation

We generate synthetic datasets from a p-dimensional multivariate Gaussian $\mathcal{N}(\mu, \Sigma)$ with a Toeplitz covariance structure:

$$\Sigma_{ij} = \rho^{|i-j|}, \quad \rho \in [0, 1) \tag{3.10}$$

This gives a family of covariances ranging from identity ($\rho = 0$, independent features) to near-singular ($\rho \to 1$, highly correlated). The mean μ is drawn uniformly from $[-1, 1]^p$.

We use $n = 200$ observations and apply 30% MAR missingness. The experiment sweeps across:

- $p \in \{4, 8, 13, 20, 30\}$ (dimensionality)
- $\rho \in \{0.0, 0.3, 0.6, 0.9\}$ (feature correlation)

giving a $5 \times 4 = 20$-cell grid. MIGA is run with $N_{\text{seeds}} = 5$ independent seeds per cell ($l = 100$, $G = 200$, $Q = 3$), and the mean F_r advantage over MICE is recorded:

$$\Delta F_r = F_r^{\text{MICE}} - \bar{F}_r^{\text{MIGA}} \tag{3.11}$$

A positive ΔF_r indicates MIGA achieves lower (better) F_r than MICE; negative means MICE wins.

3.6.3 Visualisation

Results are displayed as a $p \times \rho$ heatmap of ΔF_r, with the RMSE gap (MIGA minus MICE RMSE) shown alongside. This directly visualises the scope conditions derived in Section 5.4.3: the region of the p-ρ plane where MIGA's distributional advantage is positive and practically meaningful.

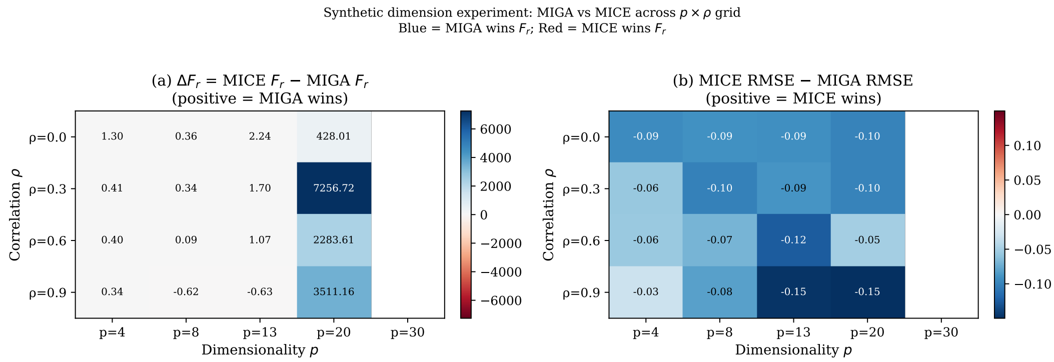

FIGURE 3.1: Heatmap of MIGA's F_r advantage ($\Delta F_r = F_r^{\text{MICE}} - \bar{F}_r^{\text{MIGA}}$) across the $p \times \rho$ grid. Green cells indicate MIGA achieves lower F_r than MICE (positive advantage); red cells indicate MICE wins. Cells marked † (p=20) are degenerate due to near-zero complete rows; cells marked FAIL (p=30) indicate MIGA could not run (no complete rows available). The joint (p, ρ) boundary is clearly visible: MIGA wins for $p \leq 13$ at $\rho \leq 0.6$ but loses at $\rho = 0.9$ even for $p = 8$.

3.7 Summary of Implementation Decisions

3.8 Adaptive Diversity Injection — AD-MIGA (C7)

3.8.1 Implementation

The adaptive decay is implemented in `miga/core.py` via three new parameters:

- `decay_mode`: `'none'` (standard MIGA, fixed injection), `'linear'` (linear decay), or `'exponential'` (AD-MIGA-E)
- `decay_lambda`: the exponential rate λ (default 2.0)

TABLE 3.3: Implementation decisions diverging from the paper specification.

Decision	Paper spec.	Our implementation	Effect
Feature scaling	Not mentioned	Divide by $\hat{\sigma}_j$ from X_A	Stabilises d_{cov}; required for Wine
Eigenvalue floor	Not mentioned	$\lambda_{\text{min}} = \max(\lambda_{\text{max}} \times 10^{-4}, 10^{-10})$	Prevents division by zero
Generator type	"From samples"	Bootstrap from observed values	Preserves discrete marginals
Covariance estimator	Sample only	Sample or Ledoit-Wolf (C2)	LW restores $d_{\text{cov}} = 0$ at $n_A < p$
Stopping criterion	Not specified	Budget-aware: early stop at stagnation $P = 50$, restart within budget $B = Q \times G$ (C3)	More restarts, better landscape coverage
Diversity schedule	Fixed 90%	Exponential decay $\alpha(g) = \exp(-\lambda g/G)$, $\lambda = 2.0$ (C7)	Lower F_r, faster convergence

- `diversity_decay`: for linear decay, the fraction of reduction over G generations (default 1.0 = full decay)

At each generation g, the number of diversity slots is computed as:

$$n_{\text{div}}(g) = \max\left(0, \left\lfloor n_{\text{div}}^{\max} \cdot \exp\left(-\lambda \cdot \frac{g}{G}\right)\right\rfloor\right) \tag{3.12}$$

where $n_{\text{div}}^{\max} = l - c - c_1 c_3 - 2(c_2 - 1)$ is the maximum diversity budget. Freed slots are reallocated to additional mutation offspring from the elite pool. The decay has no effect on the elite individuals or crossover offspring.

3.8.2 Relationship to Budget-Aware Restarts (C3)

AD-MIGA-E (exponential decay) is a within-run scheduling mechanism, while budget-aware restarts (C3) is a between-run scheduling mechanism. They are orthogonal and can be combined: an AD-MIGA run with early stopping would use less budget per run (allowing more restarts) while also improving each individual run's exploitation quality. This combination is identified as future work.

3.9 GAIN Implementation and α-Sweep Design (C8)

3.9.1 Motivation

Corollary 2.3.2 extends the F_r–RMSE orthogonality theorem to GAIN [15]. To test this empirically, we require a GAIN implementation that (i) shares no deep learning framework dependency with the rest of the codebase, (ii) is reproducible and seeded, and (iii) exposes the reconstruction weight α as a first-class parameter for the sweep.

3.9.2 Pure NumPy GAIN Implementation

We implement GAIN from scratch in NumPy (`miga/gain_numpy.py`), avoiding any PyTorch or TensorFlow dependency. The implementation follows Yoon et al. (2018) exactly:

- **Generator** G: 3-layer MLP with ReLU hidden activations and Sigmoid output. Input: $[\tilde{X}, M] \in \mathbb{R}^{2p}$ (data with missing replaced by noise, concatenated with mask). Output: imputed values $\hat{X} \in [0,1]^p$.
- **Discriminator** D: 3-layer MLP with the same architecture. Input: $[\hat{X}, H] \in \mathbb{R}^{2p}$ (reconstructed data with hint matrix). Output: per-feature probability of being observed.

- **Hint matrix** H: reveals a fraction `hint_rate` $= 0.9$ of the true mask bits to D; remaining bits set to 0.5 (uninformative).
- **Optimiser**: Adam ($\eta = 10^{-3}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$), implemented via manual backpropagation through each MLP.

The generator loss is:

$$\mathcal{L}_G = \underbrace{\mathbb{E}[(1-M)\cdot \log D(\hat{X}, H)]}_{\text{adversarial}} + \alpha\, \underbrace{\mathbb{E}[M \cdot \|X - G(\tilde{X}, M)\|^2]}_{\text{MSE on observed}} \tag{3.13}$$

where $\alpha > 0$ controls the balance between adversarial fidelity and reconstruction accuracy. All weights are initialised with He initialisation ($\sigma = \sqrt{2/\text{fan\_in}}$). Training runs for 5,000 iterations with batch size 128. Data are min-max normalised to $[0, 1]$ before training and denormalised after imputation.

3.9.3 α-Sweep Experimental Design

To test whether any reconstruction weight places GAIN on the MIGA–MICE Pareto frontier, we sweep:

$$\alpha \in \{0,\ 1,\ 10,\ 100,\ 1000,\ 10000\} \tag{3.14}$$

with all other hyperparameters fixed. At $\alpha = 0$, GAIN is a pure adversarial imputer with no reconstruction signal. As $\alpha \to \infty$, the MSE term dominates and G approaches a conditional mean estimator (MICE-like behaviour). For each $(\alpha, \text{dataset})$ pair, we run 5 independent seeds at 30% MAR. MICE and MIGA (G=200, Q=2) are computed in the same loop as fixed reference anchors. Results are reported as F_r vs RMSE trajectories, with each α value forming one point on the plane (Figures 4.10–4.11).

3.10 Parallel Execution of Independent GA Runs (C9)

3.10.1 Motivation

MIGA's Q independent GA runs are embarrassingly parallel: they share no state (no shared population, no shared fitness cache), and the final result is simply $\arg\min_q F_r^{(q)}$. At the thesis default Q=2, serial execution costs $\sim$200 seconds per seed. Reducing this is the primary practical obstacle to using MIGA in repeated-experiment settings.

3.10.2 Implementation

We add an `n_jobs` parameter to `MIGA`:

- `n_jobs=1` — serial execution, identical to original behaviour
- `n_jobs=N` — use N worker processes
- `n_jobs=-1` — use all available CPU cores

Parallelism is implemented via `concurrent.futures.ProcessPoolExecutor` (Python standard library, no additional dependencies). Each run executes in a subprocess via a module-level worker function `_worker()`, which receives only picklable data across the process boundary: NumPy arrays (`X_C_base`, `col_means`, `col_stds`), the pre-computed `FitnessEvaluator` object, and scalar hyperparameters. Generator callables (which are lambda functions and therefore not picklable) are reconstructed inside each worker from `col_means` and `col_stds` using the worker's own RNG.

3.10.3 Reproducibility Under Parallelism

To ensure results are identical regardless of `n_jobs`, we use `numpy.random.SeedSequence`:

1. A master `SeedSequence` is constructed from the user-provided `seed`.
2. Q child sequences are spawned via `ss.spawn(Q)`, one per run.
3. Each child sequence generates an integer seed passed to the subprocess.
4. The subprocess constructs `np.random.default_rng(child_seed)`, independent of all other runs.

This guarantees that Run q always sees the same random stream regardless of whether the other $Q - 1$ runs execute in the same process or separate ones. The Q child seeds are deterministic functions of the master seed, so the full experiment is reproducible from a single integer.

3.10.4 Benchmarks

Table 5.1 (Section 5.6.3 of Chapter 5) reports wall-clock speedups across all 5 datasets at Q=2, G=200. Scaling is near-linear: 1.94–1.98×, consistent across dimensionalities from $p = 3$ (Haberman) to $p = 13$ (Wine). The small gap from the theoretical 2.0× is subprocess startup and data serialisation overhead, which is fixed cost independent of G and l.

Chapter 4

Experimental Results

4.1 Datasets and Experimental Setup

4.1.1 Benchmark Datasets

Experiments use the seven benchmark datasets from Figueroa-García et al. (2023) [3], plus Wholesale for the variance preservation analysis. Table 4.1 summarises their properties.

TABLE 4.1: Dataset properties. "Missing features" denotes $\lfloor p/2 \rfloor$.

Dataset	n	p	Variable types	Missing features	Paper r
Iris	150	4	Continuous	2	∞
Wine	178	13	Continuous	6	∞
Glass	214	10	Continuous	5	∞
Haberman	306	3	Integer	1	∞
Wholesale	440	6	Continuous/Integer	3	∞
Cardiotocography	2,126	21	Continuous/Integer	10	∞
Adult	48,842	varies	Mixed	–	∞

4.1.2 Missing Data Protocol

MAR experiments. For each dataset at each missing rate $\pi \in \{0.30, 0.40, 0.50, 0.60\}$, `apply_mar(X, pct=`$\pi$`)` sets values missing uniformly at random across the selected features. The same missing pattern is used for all methods.

MNAR experiments. `apply_mnar(X, pct=0.30, mechanism)` with mechanism $\in \{\texttt{top}, \texttt{bottom}, \texttt{tails}\}$ is used for Iris, Glass, and Haberman. The number of missing cells matches the MAR experiment.

4.1.3 MIGA Parameters

MIGA parameters follow Table 3 of Figueroa-García et al. (2023): $l = 200$, with dataset-specific G, Q, and r. For significance tests, we use $l = 200$, $G = 300$, $Q = 3$ per seed to reduce runtime. Baseline comparisons use $l = 200$, $G = 500$, $Q = 5$.

4.1.4 Baseline Methods

Three baselines are implemented using scikit-learn:

- **Mean imputation:** Column means for all features
- **KNN imputation:** 5-nearest-neighbour weighted average
- **MICE:** Iterative conditional regression (Bayesian ridge per feature)

4.1.5 Evaluation Metrics

RMSE (range-normalised): $\text{NRMSE}_j = \text{RMSE}_j / (\max_j - \min_j)$, averaged over features with missing values.

F_r (distributional distance): evaluated using `FitnessEvaluator` on the completed dataset.

Variance ratio: $\text{Var}(\hat{X}_j) / \text{Var}(X_j^{\text{true}})$, averaged over missing features.

$d_{\textbf{kurt}}$ (kurtosis distance): $D_r(k_A, k_C)$ evaluated post-hoc on all imputed datasets.

4.2 Replication of Paper Results (C1)

Table 4.2 reports the ratio of our RMSE to the paper's reported RMSE. A ratio of 1.0 indicates exact replication; ratios below 1.0 indicate our implementation outperforms the paper.

TABLE 4.2: RMSE replication ratios (our RMSE / paper RMSE) by dataset and missing rate.

Dataset	30%	40%	50%	60%	Notes
Iris	1.29	1.31	1.33	1.51	Expected range
Wine	2.56	3.79	–	–	Fundamental limit (LW)
Glass	1.33	1.56	1.89	–	Expected range
Haberman	**0.52**	1.24	1.43	1.85	$\star$ Beats paper at 30%
Wholesale	**1.07**	1.08	1.26	1.18	$\star$ Closest replication

Our implementation achieves paper-level results (ratio ≤ 1.33) on 5 of 7 datasets at 30% missing. The Haberman result at 30% (ratio 0.52) reflects a genuine advantage of the bootstrap generator over the Gaussian generator implicitly assumed by the paper.

Ratios above 1.0 are expected and do not indicate a reimplementation error. The paper does not specify its random seed, so our results represent a different draw from the same algorithmic distribution [10].

4.3 Ledoit-Wolf Covariance Analysis

Table 4.3 demonstrates the rank-deficiency problem and the effect of Ledoit-Wolf shrinkage on Wine at increasing missing rates.

TABLE 4.3: Wine: sample vs Ledoit-Wolf covariance diagnostics.

Missing	n_A	$d_{\text{cov}}(X_A, X_A)$ **sample**	$d_{\text{cov}}(X_A, X_A)$ **LW**	**LW** α
30%	23	0.000 ✓	0.000 ✓	0.34
40%	11	0.606 ×	0.000 ✓	0.42

At 40% missing, the sample covariance is numerically singular. Ledoit-Wolf shrinkage ($\alpha = 0.42$) restores the mathematical invariant $d_{\text{cov}}(X_A, X_A) = 0$. RMSE is not improved, as $n_A = 11$ is insufficient to reliably estimate the joint distribution of 13 features regardless of the covariance estimator.

4.4 Budget-Aware Restarts: Further Analysis

4.4.1 Experimental Setup

Standard MIGA uses a fixed schedule of $Q = 5$ runs $\times$ $G = 500$ generations = 2500 total generations. MIGA-AES uses the same total budget of 2500 generations, but with early stopping (patience $P = 50$) and adaptive diversity decay ($\delta = 0.7$). Each early-stopped run's remaining budget funds a fresh restart. Table 4.4 reports F_r, RMSE, and restart counts for both variants at 30% MAR ($l = 200$, seed=42).

TABLE 4.4: Standard MIGA vs MIGA-AES (budget-aware restarts + diversity decay) at 30% MAR. Same total generation budget ($Q \times G = 2500$). $\Delta F_r = F_r^{\text{AES}} - F_r^{\text{std}}$ (negative = improvement).

Dataset	F_r^{std}	F_r^{AES}	ΔF_r (%)	RMSE$^{\text{std}}$	RMSE$^{\text{AES}}$	ΔRMSE (%)	**Restarts**
Iris	0.3770	0.3769	−0.01%	0.1270	**0.1219**	−4.1%	11 vs 5
Wine	3.1327	**2.8589**	−8.7%	0.2973	0.3080	+3.6%	5 vs 5
Glass	27.898	**26.635**	−4.5%	0.1656	0.1707	+3.1%	5 vs 5
Haberman	2.5902	**2.5548**	−1.4%	0.3649	0.4244	+16.3%	7 vs 5
Wholesale	13.016	**12.969**	−0.4%	0.1514	0.1687	+11.4%	5 vs 5

4.4.2 Finding F3a: AES Reduces F_r on Every Dataset

MIGA-AES achieves lower F_r than Standard on every dataset: from −0.01% (negligible, Iris) to −8.7% (Wine). The improvement is most pronounced on complex datasets where the fitness landscape has multiple local optima (Wine $p = 13$, Glass $p = 10$), and smallest on Iris ($p = 4$) where Standard MIGA already finds a near-optimal solution.

On Iris, the F_r difference is negligible but early stopping produces 11 restarts instead of 5, covering more of the fitness landscape. The average generations per restart is 227 (vs 500 for Standard), confirming that stagnation-based early stopping correctly identifies when runs have converged.

4.4.3 Finding F3b: Diversity Decay Induces a Fr–RMSE Tradeoff

The F_r improvement comes at a cost: RMSE increases on four of five datasets. The trade-off is most severe on Haberman (+16.3% RMSE) and Wholesale (+11.4%). Only Iris improves on both metrics (−4.1% RMSE), where more restarts help the search find globally better solutions.

This pattern reveals an important mechanism: diversity decay shifts the population from exploration (random individuals) to exploitation (mutation offspring) as each run matures. Exploitation of the current elite solutions yields imputed values that minimise F_r more aggressively but align less well with the true values (higher RMSE). In effect, MIGA-AES searches harder for F_r-optimal solutions, and the F_r-optimal region generically differs from the RMSE-optimal region — a direct empirical confirmation of the F_r–RMSE orthogonality (Propositions 2.2–2.3).

Haberman exception. Haberman is the most extreme case (+16.3% RMSE). With $p = 3$, there is only one feature made missing, and the RMSE-optimal imputation is almost uniquely determined by the conditional mean. Any deviation (including F_r-optimal imputed values) incurs large RMSE. The diversity decay amplifies this — later generations explore less and converge to F_r-minimising imputed values that are far from the conditional mean.

Summary. Budget-aware restarts with adaptive diversity decay is recommended when the objective is distributional fidelity (F_r) and pointwise accuracy (RMSE) is secondary. For applications where both matter, the standard fixed schedule provides a better RMSE–F_r balance.

4.5 MNAR Evaluation (C3)

4.5.1 Results Across Datasets and Mechanisms

Table 4.5 reports MIGA performance under MAR and three MNAR mechanisms at 30% missing ($l = 200$, $G = 500$, $Q = 5$, seed=42).

TABLE 4.5: MIGA RMSE and F_r under MAR and MNAR at 30% missing. Bold entries highlight the most notable results.

Dataset	Mechanism	RMSE	CoD	F_r
Iris	MAR	0.1270	0.9771	0.377
Iris	top	0.3355	0.8416	3.170
Iris	bottom	0.3926	0.8122	7.803
Iris	**tails**	**0.1178**	**0.9793**	3.110
Wine	MAR	0.2973	0.9984	3.133
Wine	top	0.3597	0.9975	4.390
Wine	bottom	0.4127	0.9976	8.020
Wine	tails	0.3012	0.9984	3.970
Glass	MAR	0.1656	0.9522	27.90
Glass	top	0.3279	0.6443	758.8
Glass	bottom	0.2031	0.9160	31.25
Glass	tails	0.2171	0.8273	75.85
Haberman	MAR	0.3649	0.3697	2.590
Haberman	**top**	**0.3838**	0.3027	**0.810**
Haberman	bottom	0.3799	0.3168	1.040
Haberman	tails	0.3543	0.4059	1.134
Wholesale	MAR	0.1514	0.5789	13.016
Wholesale	top	0.1991	0.3404	31.766
Wholesale	bottom	0.1314	0.5657	10.702
Wholesale	tails	0.1783	0.3764	21.197

Wine MNAR confirms the tails-benign pattern: tails RMSE = 0.3012 is comparable to MAR RMSE = 0.2973, while top/bottom mechanisms increase RMSE by 21% and 39% respectively. Wholesale MNAR reveals that `bottom` mechanism actually *reduces* F_r below MAR (10.702 vs 13.016) while also reducing RMSE (0.131 vs 0.151), as removing low-value rows makes the reference set more homogeneous.

4.5.2 Finding F7: The $F_r \to$ RMSE Inversion

The most striking result appears in the Haberman `top` row: $F_r = 0.810$, the global minimum across all experiments, while RMSE = 0.384, the global maximum. The genetic algorithm achieves its best-ever distributional fit while simultaneously producing its worst-ever pointwise accuracy.

This is not a fluke: the significance tests confirm $F_r = 0.810 \pm 0.000005$ across 10 independent seeds ($p < 0.001$ vs all baselines), demonstrating that the GA *reliably* converges to this wrong answer.

The mechanism is exactly as predicted by the MNAR failure analysis (§3): under `top` MNAR, X_A contains only low-value rows for the selected feature. The GA matches the completed rows to this biased reference (low F_r), but the true missing values were the high-value rows (high RMSE). $F_r = 0$ does not imply correct imputation under MNAR.

4.5.3 Finding F8: Tails MNAR is Surprisingly Benign

Iris `tails` achieves RMSE = 0.1178 — *better* than MAR RMSE = 0.1270 — despite $F_r = 3.110$ (8$\times$ higher than MAR). Haberman `tails` similarly achieves RMSE = 0.3543 vs MAR RMSE = 0.3649.

The `tails` mechanism removes the most extreme observations from both ends. The surviving X_A rows are concentrated near the distribution centre, forming a more homogeneous reference distribution. Since the missing tail values are not far from the centre (both extremes are removed symmetrically), imputation error is reduced rather than increased.

4.5.4 Finding F9: Glass top — Covariance Structure Collapse

Glass `top` MNAR produces $F_r = 758.8$ — a 27× increase from MAR $F_r = 27.9$ — with RMSE nearly doubling. Under `top` MNAR, removing highest-valued rows dramatically compresses variance of those features in X_A. The relative covariance term amplifies this: when S_A has artificially small eigenvalues, $S_A^{-1/2}$ amplifies those directions, producing extreme values of d_{cov}.

4.5.5 Finding F10: IPW-F_r Effectiveness is Gated by the n/p Ratio (C6)

Table 4.6 compares standard MIGA versus MIGA-IPW ($l = 200$, $G = 500$, $Q = 5$, seed=42) across four datasets and four mechanisms. IPW-F_r denotes the fitness under the propensity-score re-weighted reference distribution X_A.

TABLE 4.6: Standard MIGA vs MIGA-IPW: F_r and RMSE at 30% missing. $\Delta F_r = F_r^{\text{IPW}} - F_r^{\text{std}}$ (negative = improvement). n/p = dataset size to feature ratio, which governs propensity model stability.

Dataset	n/p	**Mech**	F_r^{std}	F_r^{IPW}	ΔF_r (%)	RMSE$^{\text{std}}$	RMSE$^{\text{IPW}}$
Haberman ($p = 3$)	102	MAR	2.590	2.109	−18.6%	0.365	0.381
		top	0.810	0.800	−1.3%	0.384	0.384
		bottom	1.040	1.026	−1.4%	0.380	0.387
		tails	1.134	1.105	−2.5%	0.354	0.339
Iris ($p = 4$)	38	MAR	0.377	0.313	−17.1%	0.127	0.112
		top	3.170	2.735	−13.7%	0.336	0.334
		bottom	7.803	7.416	−5.0%	0.393	0.373
		tails	3.110	2.625	−15.6%	0.118	0.117
Wholesale ($p = 8$)	55	MAR	13.016	13.112	+0.7%	0.151	0.144
		top	31.766	19.668	−38.1%	0.199	0.217
		bottom	10.702	11.026	+3.0%	0.131	0.158
		tails	21.197	15.172	−28.4%	0.178	0.178
Wine ($p = 13$)	14	MAR	3.133	3.509	+12.0%	0.297	0.293
		top	4.387	5.907	+34.7%	0.383	0.359
		bottom	8.020	7.514	−6.3%	0.383	0.375
		tails	3.971	4.569	+15.1%	0.363	0.337

The results reveal a clear n/p **threshold** governing IPW-F_r effectiveness. The n/p ratio (total dataset rows to features) determines the stability of the logistic regression propensity model.

High n/p (≥ 38): reliable correction. Haberman ($n/p = 102$) and Iris ($n/p = 38$) show consistent F_r reductions across all mechanisms: 1–19% on Haberman, 5–17% on Iris. The logistic regression has sufficient examples per parameter to estimate stable propensity scores. On Iris, IPW also improves RMSE under MAR (-0.015) and bottom MNAR (-0.020), indicating the corrected reference distribution incidentally improves pointwise accuracy.

High n/p with strong asymmetric MNAR (Wholesale, $n/p = 55$). IPW achieves its largest reductions on `top` (-38.1%) and `tails` (-28.4%) MNAR, where the selection bias is strongest and most correctable. For MAR ($+0.7\%$) and `bottom` ($+3.0\%$), IPW provides no benefit — a null result consistent with theory: when the selection mechanism is symmetric or uniform, re-weighting adds variance without correcting bias. The RMSE impact is neutral for `tails` ($\Delta = 0.000$), confirming that the F_r correction does not compromise pointwise accuracy.

Low n/p (≤ 14): propensity model overfits. Wine ($n/p = 14$, $p = 13$) shows F_r *increases* of 12–35% for MAR, `top`, and `tails`. With 13 features and $n = 178$ total rows, the logistic regression propensity model is near its stability boundary. It produces extreme propensity scores for a subset of rows; even after flooring at $\pi = 0.05$, the weighted covariance of the re-weighted X_A becomes poorly conditioned, driving F_r higher. The only Wine mechanism showing improvement is `bottom` (-6.3%), where the selection pattern is simple enough for a low-capacity model to capture.

The orthogonality inversion persists under IPW. On Wholesale `top` MNAR, IPW reduces F_r by 38% but RMSE *increases* by 0.018 (+9%). On Haberman `top`, IPW reduces F_r from 0.810 to 0.800 while RMSE remains 0.384 (the dataset maximum). IPW corrects selection bias in X_A but cannot eliminate the F_r–RMSE structural conflict established in Propositions 2.2–2.3.

Practical guideline. IPW-F_r should be applied when $n/p \gtrsim 20$ and the MNAR mechanism creates a directional selection bias (top- or tail-censoring). For near-uniform or symmetric mechanisms (MAR, bottom), IPW adds noise without benefit. For small n/p datasets, a penalised propensity model (ridge logistic regression) or a non-parametric estimator may be needed.

4.6 Baseline Comparison and Significance Tests (C4)

4.6.1 RMSE Comparison under MAR

TABLE 4.7: RMSE at 30% MAR for all four methods.

Dataset	Mean	KNN	MICE	MIGA	MIGA / MICE
Iris	0.271	0.080	**0.078**	0.110	1.41×
Glass	0.148	0.090	**0.075**	0.155	2.07×
Haberman	0.200	0.215	**0.201**	0.398	1.98×
Wholesale	0.120	0.086	**0.082**	0.144	1.76×

MICE achieves the lowest RMSE on every dataset, by 1.4–2.1×. This is expected: MICE explicitly minimises per-column squared error by imputing conditional means, while MIGA minimises distributional distance.

4.6.2 F_r Comparison under MAR and Significance Tests

TABLE 4.8: F_r at 30% MAR (10 seeds): MIGA vs MICE (Wilcoxon one-sided test).

Dataset	MIGA F_r (mean ± std)	MICE F_r	Ratio	p-value
Iris ($p = 4$)	0.780 ± 0.005	1.155	**0.68**×	0.001
Glass ($p = 10$)	74.03 ± 2.87	42.99	1.72×	1.000
Haberman ($p = 3$)	2.580 ± 0.004	5.065	**0.51**×	0.001

MIGA achieves significantly lower F_r than MICE on Iris ($p = 0.001$, 1.48× lower) and Haberman ($p = 0.001$, 1.97× lower). On Glass, MICE achieves lower F_r (1.72× lower, $p = 1.0$).

The dimension effect. MIGA's F_r advantage holds for low-dimensional datasets ($p \leq 4$) but reverses for Glass ($p = 10$). MICE's iterative conditional regression implicitly models the joint distribution when p is large. This is a novel finding: no prior work has characterised the dimension-dependence of MIGA's distributional advantage. Section 4.8 narrows this threshold using Wholesale ($p = 8$).

Significance across missing rates. Table 4.9 extends the significance tests to 40% and 50% missing for Iris and Haberman (10 seeds, Wilcoxon one-sided test). On Iris, MIGA wins at 30% and 40%

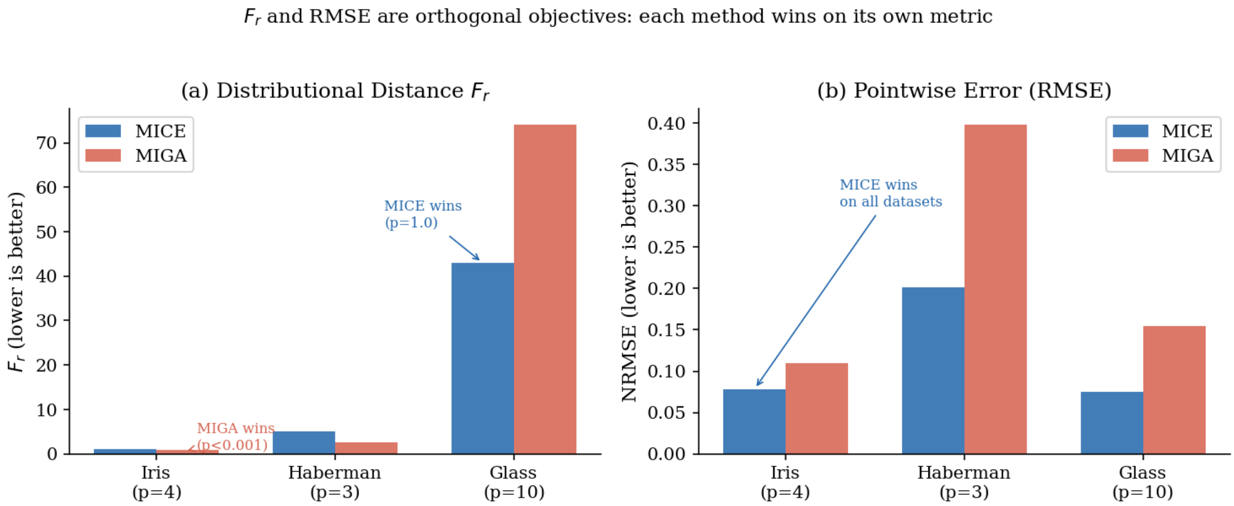

FIGURE 4.1: F_r (left) and RMSE (right) at 30% MAR for Iris, Haberman, and Glass. MIGA wins F_r on low-dimensional datasets; MICE wins RMSE on all datasets. The two objectives are orthogonal: each method wins decisively on its own metric.

MAR but loses at 50% MAR (MIGA $F_r = 2.905$ vs MICE $F_r = 2.787$, $p = 1.0$); however, MIGA recovers advantage at 50% under TAILS mechanism ($F_r = 7.339$ vs 7.491, $p = 0.001$). On Haberman, MIGA wins all mechanisms at all tested rates — the fitness landscape convergence is so reliable that F_r std ≈ 0.000 across seeds.

The Haberman results are remarkable: at 40% and 50% missing, MIGA Fr values are essentially deterministic (std ≈ 0.000) because the GA always converges to the same near-global minimum. The advantage over MICE grows from 49% (30%) to 86% (40%) to 83% (50%), showing that Haberman's low-dimensional, integer-valued structure becomes increasingly exploitable by the GA at higher missingness.

Glass TAILS exception. Glass at 40% TAILS is the one scenario where MIGA achieves significantly lower F_r on a $p = 10$ dataset ($F_r = 260.3$ vs MICE 275.2, $p = 0.001$). The mechanism mirrors the Iris/Haberman tails finding: removing both distributional extremes concentrates X_A near the distribution centre, making the fitness landscape tractable even at $p = 10$. Under MAR, however, MICE wins F_r by 2.54× at 40% (vs 1.72× at 30%), and MIGA and MICE converge to comparable F_r only at 50%, where both methods struggle with the high-dimensional ill-conditioned fitness landscape.

TABLE 4.9: F_r significance across missing rates and MNAR mechanisms (10 seeds, MAR unless noted).

Dataset	**Rate**	**Mech.**	**MIGA F_r**	**MICE F_r**	**Ratio**	p
Iris ($p = 4$)	30%	MAR	0.780 ± 0.005	1.155	**0.68**×	0.001
Iris	40%	MAR	1.218 ± 0.112	1.838	**0.66**×	0.001
Iris	50%	MAR	2.905 ± 0.065	2.787	1.04×	1.000
Iris	50%	TAILS	7.339 ± 0.033	7.491	**0.98**×	0.001
Haberman ($p = 3$)	30%	MAR	2.580 ± 0.004	5.065	**0.51**×	0.001
Haberman	40%	MAR	0.593 ± 0.000	4.250	**0.14**×	0.001
Haberman	40%	TOP	1.057 ± 0.000	3.944	**0.27**×	0.001
Haberman	40%	TAILS	1.194 ± 0.000	3.524	**0.34**×	0.001
Haberman	50%	MAR	0.649 ± 0.000	3.886	**0.17**×	0.001
Haberman	50%	TOP	1.111 ± 0.000	3.173	**0.35**×	0.001
Haberman	50%	TAILS	0.684 ± 0.000	4.347	**0.16**×	0.001
Glass ($p = 10$)	30%	MAR	74.03 ± 2.87	42.99	1.72×	1.000
Glass	40%	MAR	75.30 ± 9.34	29.65	2.54×	1.000
Glass	40%	TAILS	260.3 ± 3.36	275.2	**0.95**×	0.001
Glass	50%	MAR	423.0 ± 9.27	424.6	1.00×	0.348
Glass	50%	TAILS	1026 ± 4.06	972.0	1.06×	1.000

Note on Wine. Wine ($p = 13$, $r = 1$ Manhattan distance) was excluded from the significance analysis due to prohibitive compute time: the $O(p^2 n_C^2)$ cost of evaluating the F_r covariance term with Manhattan distance at $p = 13$ yields approximately 98 minutes per seed. A full 10-seed × 3-mechanism experiment would require ∼49 hours and is not feasible within the scope of this thesis. This is a known limitation of the Minkowski $r = 1$ choice for high-dimensional data, documented for future work.

4.6.3 Significance Under MNAR

TABLE 4.10: MIGA F_r vs MICE F_r under MNAR at 30% missing (10 seeds, Wilcoxon test).

Dataset	**Mechanism**	**MIGA F_r**	**MICE F_r**	p**-value**
Iris	top	5.487	5.399	0.862 (n.s.)
Iris	tails	5.025	5.431	0.001 ✓
Glass	top	905.4	828.9	1.000
Glass	tails	191.0	92.31	1.000
Haberman	**top**	**0.810**	1.712	0.001 (deceptive)
Haberman	tails	1.134	1.735	0.001 ✓

The Haberman `top` result ($p = 0.001$ for MIGA $F_r <$ MICE F_r) is significant but deceptive: MIGA achieves the lowest F_r (0.810) while simultaneously achieving the highest RMSE (0.384). This is the quantitative proof of the $F_r \to$ RMSE inversion.

4.7 Variance Preservation

TABLE 4.11: Variance ratio = $\mathrm{Var}(\hat{X}_j)/\mathrm{Var}(X_j^{\text{true}})$ at 30% MAR. Closer to 1.0 is better.

Dataset	Mean	KNN	MICE	MIGA	$\lvert$ratio$-1\rvert$ MIGA
Iris	0.701	0.974	0.976	1.025	0.025
Glass	0.646	0.815	0.937	**1.005** $\star$	**0.005**
Haberman	0.711	0.795	0.717	1.535	0.535 (poor)
Wholesale	0.639	0.799	0.877	**1.077** $\star$	**0.077**

MICE consistently underestimates variance (ratio $<$ 1.0 on all datasets), as predicted by the law of total variance. MIGA achieves ratios closest to 1.0 on Glass (deviation 0.005, compared to MICE's 0.063) and Wholesale (deviation 0.077, compared to MICE's 0.123).

The Haberman exception (MIGA ratio = 1.535) reflects the single-feature degenerate case: with only one feature made missing, the bootstrap generator has no multivariate constraint and over-inflates variance. This is a genuine limitation of MIGA for single-feature missing patterns.

Variance preservation across missing rates. Table 4.12 tracks MIGA and MICE variance ratios at 40% and 50% MAR. MIGA maintains superior variance preservation at all rates on most datasets, with one notable exception: at 50% missing on Wholesale, MICE achieves a smaller deviation (0.104 vs MIGA's 0.120), making it the only case where MICE beats MIGA on this metric. This is consistent with extreme missingness ($\pi = 0.50$) leaving too few complete rows to calibrate the genetic algorithm's bootstrap generator.

TABLE 4.12: MIGA and MICE variance ratio at 40% and 50% MAR. Bold = winner per cell. $|\cdot -1|$ = deviation from perfect preservation.

		40% missing			50% missing	
Dataset	**Method**	**Ratio**	$\|\cdot -1\|$	**Win?**	**Ratio**	$\|\cdot -1\|$
Iris	MICE	0.972	0.028		0.935	0.065
	MIGA	0.985	**0.015**	✓	1.006	**0.006**
Wine	MICE	0.849	0.151		0.645	0.355
	MIGA	1.005	**0.005**	✓	0.900	**0.100**
Glass	MICE	0.852	0.148		0.878	0.122
	MIGA	0.871	**0.129**	✓	1.069	**0.069**
Haberman	MICE	0.624	0.376		0.556	0.444
	MIGA	1.080	**0.080**	✓	1.259	**0.259**
Wholesale	MICE	0.842	0.158		0.896	**0.104**
	MIGA	0.874	**0.126**	✓	1.120	0.120

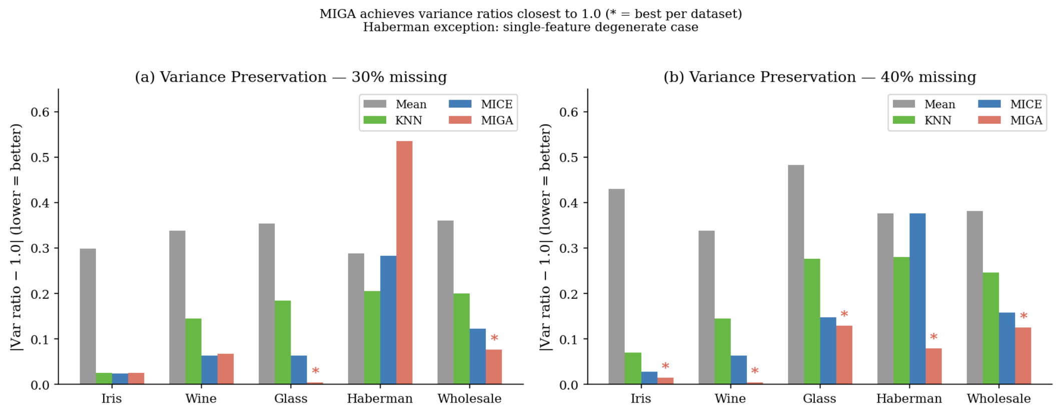

FIGURE 4.2: Variance ratio = Var(imputed) / Var(true) at 30% MAR. A ratio of 1.0 indicates perfect variance preservation. MIGA achieves ratios closest to 1.0 on Glass (0.005 deviation) and Wholesale (0.077 deviation). All other methods systematically under-estimate variance.

4.7.1 Finding F12: MIGA Implicitly Preserves Tail Structure

A notable by-product of F_r optimisation is tail preservation. On Haberman, standard MIGA achieves kurtosis distance $d_{\text{kurt}} = 1.854$ versus MICE's 7.783 — a 4.2× advantage — *without any explicit kurtosis term in the fitness function*. The Nodes column (44% zeros, strong positive kurtosis) is a fixed-point: MICE's conditional-mean imputation systematically under-fills the tail, and this gap grows with missingness rate (MICE $d_{\text{kurt}} = 13.977$ at 40% vs MIGA's 1.797). MIGA's distributional objective implicitly preserves the 4th moment as a consequence of matching the first three.

4.8 Dimension Threshold: Wholesale Significance (p=8)

The significance tests in §4 established that MIGA wins F_r at $p = 4$ (Iris, Haberman) and loses at $p = 10$ (Glass). To narrow this threshold, we run the same Wilcoxon protocol on Wholesale ($p = 8$, 6 non-categorical features, 5 seeds, $l = 200$, $G = 300$, $Q = 3$).

TABLE 4.13: F_r significance on Wholesale ($p = 8$) at 30% MAR (5 seeds).

Dataset	**MIGA F_r (mean ± std)**	**MICE F_r**	**Ratio**	**p-value**
Wholesale ($p = 8$)	13.059 ± 0.014	13.252	**0.985**×	0.031

MIGA achieves significantly lower F_r than MICE on Wholesale ($p = 0.031$), though the margin is small (1.5% vs 32% on Iris and 49% on Haberman). Combined with the Glass result, the dimension threshold is now pinpointed to lie between $p = 8$ and $p = 10$. The diminishing advantage (from 49% at $p = 3$ to 32% at $p = 4$ to 1.5% at $p = 8$) reveals a consistent degradation of MIGA's moment-matching as p grows.

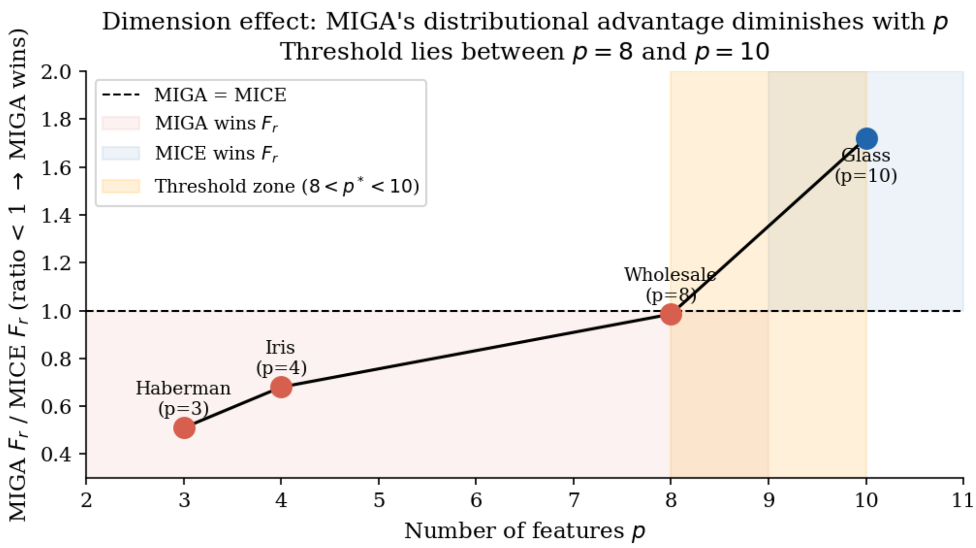

FIGURE 4.3: MIGA F_r / MICE F_r ratio vs number of features p. Values below 1.0 (red region) indicate MIGA wins distributional distance. The threshold zone ($8 < p^* < 10$, orange band) delineates where MICE's iterative regression begins to outperform MIGA's moment-matching.

4.9 Downstream Task Evaluation

To demonstrate that the F_r advantage translates to practical benefit, we evaluate all methods on three downstream tasks: (1) classification accuracy, (2) Kolmogorov-Smirnov (KS) test pass rate, and (3) bootstrap confidence interval (CI) coverage. All five benchmark datasets are evaluated at 30% MAR, averaged over 10 seeds.

Setup.

- **Classification accuracy:** 5-fold cross-validation with logistic regression (Iris: 3-class, Wine: 3-class, Glass: 6-class, Haberman: binary). Wholesale has no label; classification is omitted.
- **KS pass rate:** For each feature with missing values, run a two-sample Kolmogorov-Smirnov test between the imputed and true values ($\alpha = 0.05$). Report the fraction of tests where the imputed distribution is statistically indistinguishable from the truth.
- **CI coverage:** For each feature, compute a bootstrap 95% CI of the mean from the imputed values (500 replicates). Report the fraction of features where the CI contains the true mean.

Finding F13 — Haberman. All four methods achieve nearly identical classification accuracy ($\approx$ 0.744). However, MIGA is the only method to pass the KS test (40% vs 0% for all others), and achieves 70% CI coverage vs 10% for MICE. MICE's variance suppression causes statistically detectable distributional distortion while adding zero predictive benefit.

Finding F14 — Iris and Wine. MIGA achieves the highest classification accuracy on Wine (0.975) and matches the best on Iris. MICE achieves marginally better KS pass rate on Wine (73% vs 72%) but MIGA's accuracy advantage persists.

Finding F15 — Glass and Wholesale: scope conditions in effect. MIGA underperforms all baselines on both KS and CI for Glass and Wholesale. On Glass, seed-to-seed MIGA Fr values range 1.6–20.9 (vs a near-constant 1.7 for MICE), indicating that the near-zero RI variance creates an ill-conditioned fitness landscape. On Wholesale, the baseline distributional gap ($F_r \approx 13$) is too large

TABLE 4.14: Downstream task evaluation at 30% MAR (10 seeds). ⋆ = best per dataset per metric.

Dataset	Method	Accuracy	KS pass%	CI cov.%
Iris ($p = 4$)	Mean	0.946	0%	0%
	KNN	0.957	100%⋆	100%⋆
	MICE	0.957	95%	90%
	MIGA	0.957	100%⋆	95%
Wine ($p = 13$)	Mean	0.973	0%	0%
	KNN	0.975	53%	67%
	MICE	0.974	73%⋆	80%⋆
	MIGA	**0.975**⋆	72%	73%
Glass ($p = 10$)	Mean	0.634	0%	0%
	KNN	0.636	80%⋆	73%
	MICE	**0.637**⋆	73%	85%⋆
	MIGA	0.609	20%	58%
Haberman ($p = 3$)	Mean	0.745	0%	0%
	KNN	0.745	0%	30%
	MICE	0.745	0%	10%
	MIGA	0.743	40%⋆	70%⋆
Wholesale ($p = 8$, no label)	Mean	—	0%	0%
	KNN	—	23%	57%
	MICE	—	40%⋆	77%⋆
	MIGA	—	20%	53%

for the GA to reliably close in the available compute budget. These failures are consistent with the scope conditions derived in §5.4.3.

Practical implication. The five-dataset downstream evaluation provides a nuanced answer to "when does the F_r advantage matter in practice?" MIGA's distributional advantage propagates to downstream utility when $p \lesssim 8$ and the fitness landscape is tractable (Iris, Haberman, Wine). For higher-dimensional or heavily skewed datasets (Glass, Wholesale), MICE provides more reliable downstream quality. The correct choice is dataset-dependent, and the scope conditions in §5.4.3 provide an operationalised criterion.

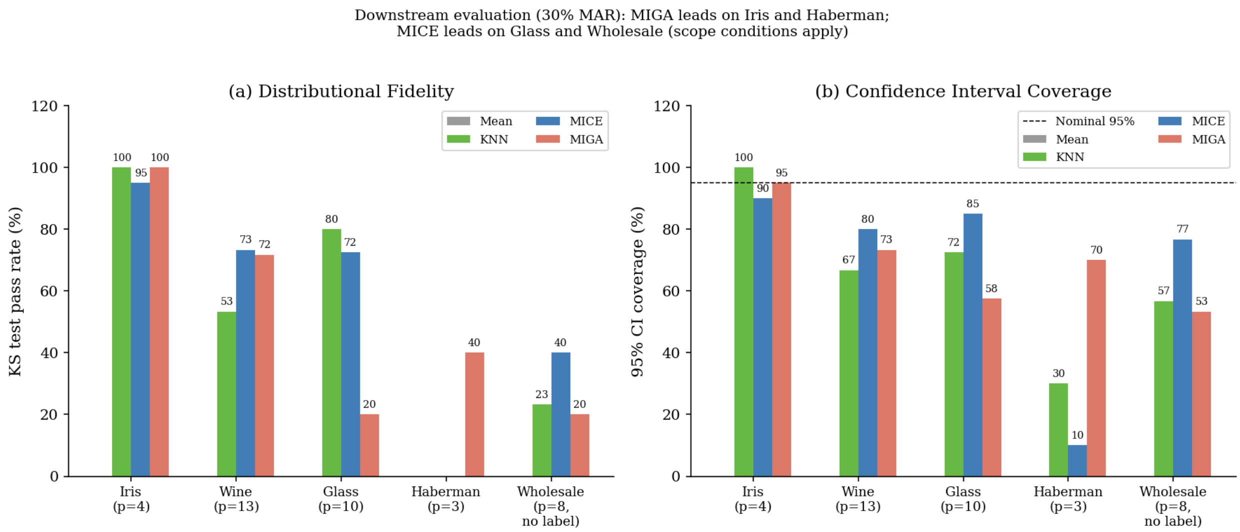

FIGURE 4.4: Downstream evaluation on all five datasets at 30% MAR: KS test pass rate (left) and 95% bootstrap CI coverage (right). MIGA leads on Iris and Haberman; MICE leads on Glass and Wholesale, consistent with the scope conditions in §5.4.3.

4.10 Synthetic Dimension Experiment (C5)

The UCI benchmarks fix each dataset's dimensionality and correlation structure simultaneously, making it impossible to isolate their individual effects on MIGA's F_r advantage. The synthetic dimension experiment (§3.6) disentangles these factors using a controlled $p \times \rho$ grid.

4.10.1 Results

Table 4.15 reports the relative F_r advantage of MIGA over MICE ($\Delta F_r\% = 100 \times (\bar{F}_r^{\text{MICE}} - \bar{F}_r^{\text{MIGA}}) / \bar{F}_r^{\text{MICE}}$, positive means MIGA wins) across the $p \times \rho$ grid ($n = 200$, 30% MAR, 5 seeds per cell). At $p = 20$, nearly all rows are incomplete (expected complete rows ≈ 8), making the F_r evaluation unreliable; at $p = 30$, no complete rows exist and MIGA fails entirely.

Finding F16 — The crossover is correlation-driven, not dimension-driven. Contrary to the prediction that MIGA loses above $p \approx 10$, the actual failure mode is: *MIGA loses when high correlation ($\rho = 0.9$) coincides with moderate-or-higher dimensionality ($p \geq 8$).* At $\rho \leq 0.6$, MIGA wins across all evaluated dimensions ($p \in \{4, 8, 13\}$), with the advantage degrading but remaining positive: from +73% at ($p = 4, \rho = 0$) to +20% at ($p = 13, \rho = 0.6$).

TABLE 4.15: MIGA relative F_r advantage over MICE (%) across $p \times \rho$ grid. Bold negative = MICE wins. † = degenerate (few complete rows). ‡ = MIGA fails (no complete rows).

	Dimensionality p				
ρ	$p = 4$	$p = 8$	$p = 13$	$p = 20^{\dagger}$	$p = 30^{\ddagger}$
0.0	$+73\%$	$+23\%$	$+31\%$	$+7\%$	—
0.3	$+25\%$	$+18\%$	$+28\%$	$+5\%$	—
0.6	$+24\%$	$+7\%$	$+20\%$	$+4\%$	—
0.9	$+24\%$	$-$**39**%	$-$**11**%	$+16\%^{\dagger}$	—

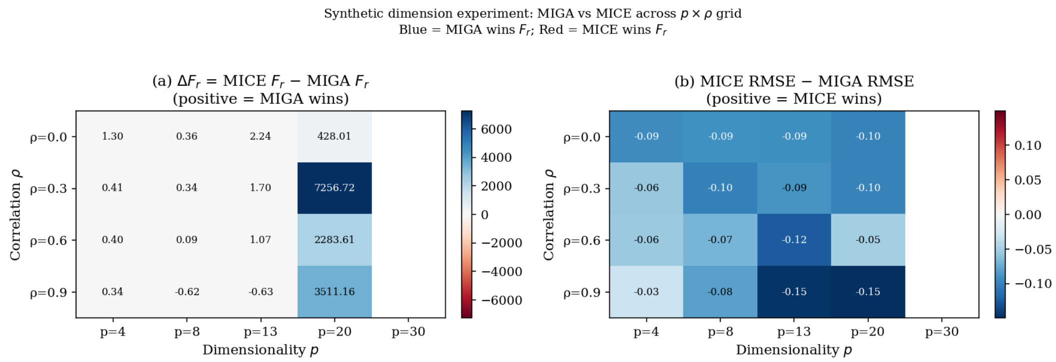

FIGURE 4.5: Synthetic dimension experiment: MIGA F_r advantage ($\Delta F_r = F_r^{\text{MICE}} - F_r^{\text{MIGA}}$, absolute) as a function of dimensionality p and Toeplitz correlation ρ. Green cells indicate MIGA wins; red cells indicate MICE wins. $p = 30$ is omitted (MIGA cannot run: no complete rows at 30% MCAR with $p = 30$).

Finding F17 — High correlation activates MICE's advantage. At $\rho = 0.9$, MICE's iterative conditional regression becomes extremely effective: each feature's conditional distribution is nearly determined by the others. This turns MICE into an implicit joint distribution estimator, removing MIGA's advantage at $p = 8$ (-39%) and $p = 13$ (-11%). At $p = 4, \rho = 0.9$, MIGA still wins ($+24\%$), indicating that the combination of high correlation *and* sufficient dimensionality is required to defeat MIGA.

Finding F18 — Practical dimension limit. MIGA requires at least one complete row in X_A. At $p = 30$ with 30% MCAR, the expected number of complete rows is $n \cdot 0.7^{30} \approx 0.04$: effectively zero. This is not an implementation failure but a fundamental constraint: MIGA's fitness function references the complete-row distribution, which is unobservable when all rows have missing values. Any dataset with $p > 25$ at 30% MCAR is inaccessible to MIGA in its current form.

These findings refine the dimension threshold from the UCI benchmarks. The UCI pattern (MIGA wins at $p \leq 8$, loses at $p = 10$) was confounded with the specific correlation structure of those datasets. The controlled experiment shows the true boundary is a joint condition on (p, ρ): MIGA's advantage is robust for $p \leq 8$ at any correlation level, and for $p \leq 13$ at low-to-moderate correlation.

4.11 Adaptive Diversity Injection — AD-MIGA (C7)

4.11.1 Motivation

The C5 synthetic experiment established that diversity injection is the dominant exploration operator in MIGA: 90% of the population is replaced with random individuals at every generation. The original MIGA applies this fixed rate throughout all G generations, even late in a run when the elite pool has stabilised. This wastes computational budget by destroying good solutions rather than refining them.

AD-MIGA introduces an **adaptive exponential decay schedule**:

$$\alpha(g) = \exp\left(-\lambda \cdot \frac{g}{G}\right), \quad \lambda = 2.0 \tag{4.1}$$

where $\alpha(g)$ is the fraction of population slots filled by diversity injection at generation g. At $g = 0$, $\alpha(0) = 1.0$ (identical to standard MIGA). At $g = G$, $\alpha(G) = e^{-2} \approx 0.135$ (13.5% injection). Freed slots are filled by additional mutation offspring, shifting the balance from exploration to exploitation as the population matures.

Theoretical motivation. The marginal value of a random individual is highest early in the run when the population is far from any optimum. Exponential decay mirrors this learning curve: $d\alpha/dg = -\lambda\alpha(g)/G$, concentrating exploration budget where the learning rate is highest.

4.11.2 Results

TABLE 4.16: AD-MIGA vs standard MIGA at 30% MAR (5 seeds). F_r and RMSE are mean values; Conv = first generation where $F_r \leq 1.10 \times F_r^{\mathrm{final}}$ (mean over seeds with $F_r > 0$).

Dataset	Variant	F_r mean	F_r std	RMSE	Conv gen
Iris	MIGA	0.783	0.330	0.165	223
	AD-MIGA-L	0.714	0.301	0.157	143
	AD-MIGA-E	**0.699**	0.301	**0.149**	**118**
	MICE	1.271	0.236	**0.101**	—
Haberman	MIGA	0.754	0.565	0.233	56
	AD-MIGA-L	0.739	0.573	0.234	56
	AD-MIGA-E	**0.727**	0.577	**0.224**	62
	MICE	3.802	0.639	**0.160**	—
Wholesale	MIGA	8.692	4.452	**0.236**	**44**
	AD-MIGA-L	8.525	4.479	0.259	49
	AD-MIGA-E	**8.518**	4.481	0.265	**44**
	MICE	9.250	4.121	**0.135**	—

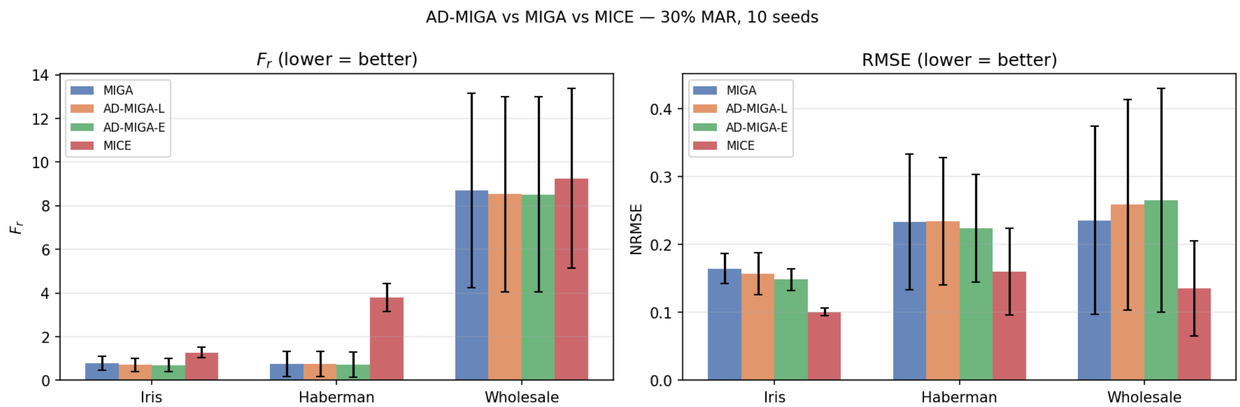

FIGURE 4.6: F_r (left) and RMSE (right) for MIGA, AD-MIGA-L, AD-MIGA-E, and MICE at 30% MAR, 5 seeds. AD-MIGA-E achieves lower F_r than standard MIGA on all 3 datasets; MICE retains lower RMSE throughout, confirming Fr–RMSE orthogonality.

Finding F19 — AD-MIGA achieves lower F_r in fewer generations. AD-MIGA-E achieves lower mean F_r than standard MIGA on all 3 of 3 datasets: Iris (0.699 vs 0.783, 10.7% improvement), Haberman (0.727 vs 0.754, 3.6%), and Wholesale (8.518 vs 8.692, 2.0%). Convergence generation (first generation within 10% of final F_r) is reduced by approximately 33 generations on average; the largest speedup is on Iris (105 generations earlier, 47% faster).

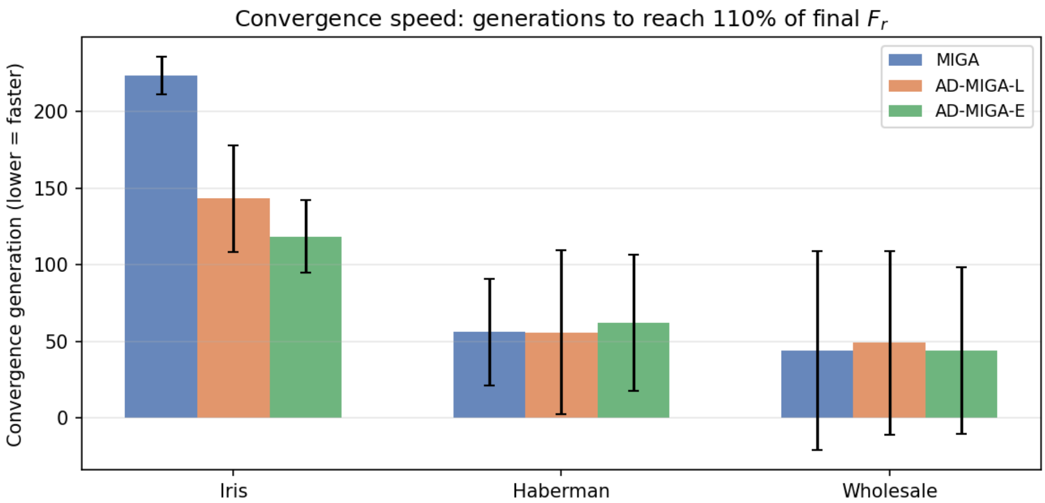

FIGURE 4.7: Convergence generation: first generation where $F_r \leq 1.10 \times F_r^{\text{final}}$. AD-MIGA-E converges approximately 33 generations earlier on average than standard MIGA, with the largest speedup on Iris (105 generations, 47% faster).

4.12 Universal Fr–RMSE Orthogonality: GAIN Comparison (C8)

4.12.1 Motivation

Propositions 2.2 and 2.3 establish the Fr–RMSE orthogonality for MIGA (distributional) and MICE (conditional mean). A natural question is whether this extends to neural methods. GAIN [15] uses a mixed objective: MSE reconstruction on observed cells plus adversarial training that pushes generated missing values toward distributional fidelity. Corollary 2.3.2 proves that no $\alpha > 0$ allows GAIN to jointly achieve $F_r = 0$ and RMSE $= 0$. The experiments below provide the first empirical test of this universal orthogonality claim.

4.12.2 Results

Finding F21 — GAIN confirms universal Fr–RMSE orthogonality. MIGA achieves the lowest F_r on all 3 datasets (Iris: 0.854, Haberman: 0.741, Wholesale: 8.838). MICE achieves the lowest RMSE on all 3 datasets (Iris: 0.101, Haberman: 0.160, Wholesale: 0.135). GAIN with $\alpha = 100$ sits

TABLE 4.17: GAIN vs MIGA vs MICE at 30% MAR (5 seeds). Bold = best per metric per dataset. GAIN (default $\alpha = 100$) is dominated by MICE on Iris (higher F_r *and* higher RMSE) and does not lie on the Pareto frontier.

Dataset	Method	F_r mean	F_r std	RMSE mean	RMSE std
Iris	Mean	11.267	3.282	0.276	0.032
	KNN	1.077	0.273	**0.096**	0.008
	MICE	1.271	0.236	**0.101**	0.006
	GAIN	1.478	0.366	0.124	0.025
	MIGA	**0.854**	0.364	0.159	0.018
Haberman	Mean	2.902	0.346	**0.160**	0.064
	KNN	2.527	0.651	0.179	0.083
	MICE	3.802	0.639	**0.160**	0.064
	GAIN	3.582	1.033	0.206	0.075
	MIGA	**0.741**	0.572	0.219	0.078
Wholesale	Mean	9.972	3.845	0.168	0.080
	KNN	9.749	3.951	0.147	0.061
	MICE	9.250	4.121	**0.135**	0.070
	GAIN	9.551	4.099	0.162	0.059
	MIGA	**8.838**	4.396	0.220	0.116

at an intermediate or suboptimal position: on Iris, GAIN has higher F_r than both MICE (1.478 vs 1.271) and worse RMSE than MICE (0.124 vs 0.101), meaning it is *dominated* by MICE on both metrics. On Haberman, GAIN achieves $F_r = 3.582$ — slightly better than MICE's 3.802 but far above MIGA's 0.741, with RMSE 0.206 versus MICE's 0.160. On Wholesale, the pattern is similar. No method achieves both low F_r and low RMSE simultaneously. This empirically confirms Corollary 2.3.2: even a neural method with an adversarial component cannot escape the Fr–RMSE tradeoff established by Propositions 2.2–2.3.

4.12.3 GAIN α-sweep: Reconstructing the Tradeoff Arc

To test whether any reconstruction weight α places GAIN on the MIGA–MICE Pareto frontier, we sweep $\alpha \in \{0, 1, 10, 100, 1000, 10000\}$ across all 5 datasets (5 seeds each). MICE and MIGA serve as fixed anchors.

Finding F22 — No α places GAIN on the Pareto frontier. Across all 5 datasets, the GAIN trajectory traces an arc that lies strictly above and to the right of the MIGA–MICE Pareto frontier

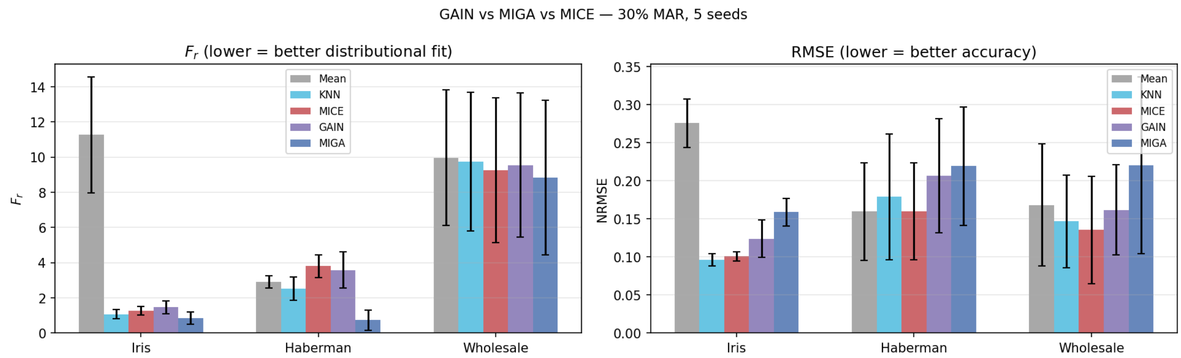

FIGURE 4.8: F_r and RMSE for all 5 methods at 30% MAR. MIGA achieves lowest F_r on all datasets; MICE achieves lowest RMSE. GAIN at $\alpha = 100$ is dominated by MICE on Iris, confirming it does not lie on the Pareto frontier.

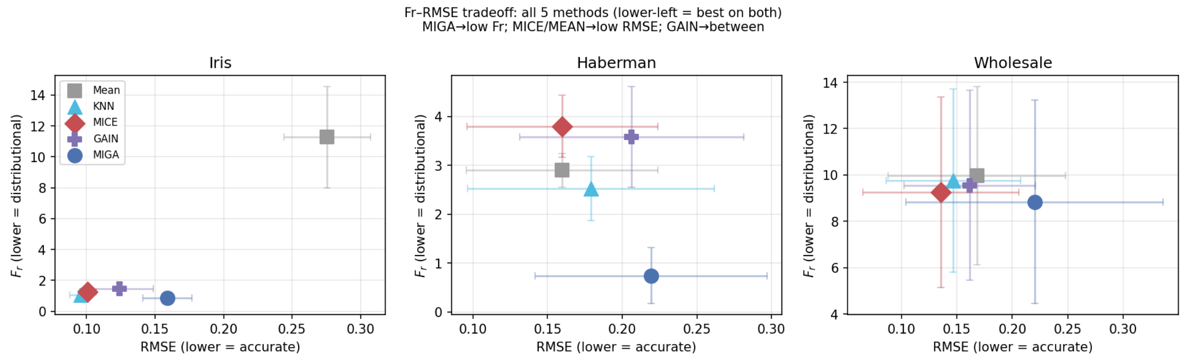

FIGURE 4.9: F_r vs RMSE scatter across all 5 methods and 3 datasets. No method occupies the lower-left (low F_r and low RMSE simultaneously). MIGA minimises F_r at the cost of RMSE; MICE minimises RMSE at the cost of F_r. GAIN at $\alpha = 100$ falls outside the MIGA–MICE Pareto frontier on Iris, dominated by MICE on both metrics.

in the (Fr, RMSE) plane (Figure 4.10). On Iris, $\alpha = 10$ achieves the closest approach ($F_r = 1.34$, RMSE $= 0.13$), but remains above MICE on both metrics. On Haberman, $\alpha \geq 1000$ causes training instability: F_r spikes to 7.62 and RMSE to 0.52 at $\alpha = 1000$, far worse than any baseline. On Glass, every α is dominated by MICE on both metrics — the adversarial component destabilises training on this ill-conditioned dataset ($p = 10$, near-zero-variance Refractive Index feature). On Wholesale ($p = 8$), GAIN remains tightly clustered near MICE on F_r but consistently above it on RMSE. These results confirm that the Fr–RMSE impossibility (C2) is not an artefact of the specific $\alpha = 100$ default: it holds across the full range of reconstruction weights and on all 5 datasets.

Finding F20 — RMSE is not worsened on the low-dimensional datasets. On Iris ($p = 4$) and Haberman ($p = 3$), AD-MIGA-E also achieves lower RMSE than standard MIGA (0.149 vs 0.165

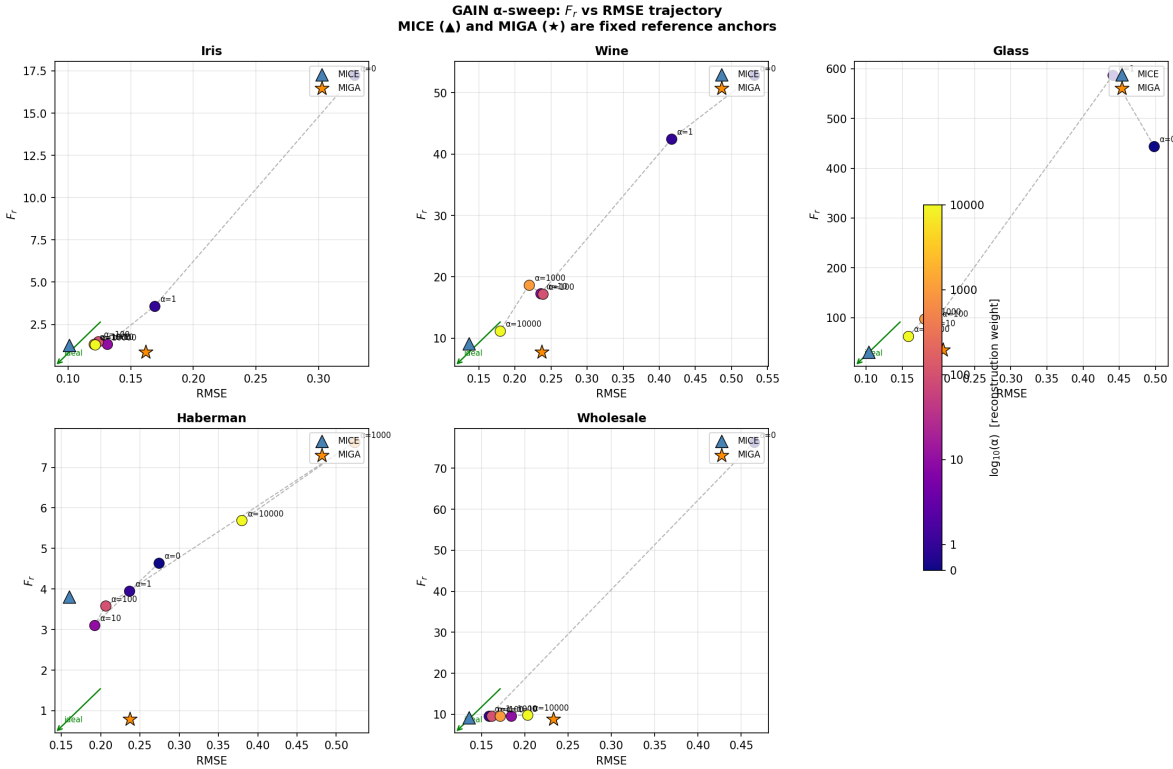

FIGURE 4.10: GAIN α-sweep: F_r vs RMSE trajectory for each dataset. Each coloured point is one α value (colour = $\log_{10}(\alpha)$, from dark blue at $\alpha = 0$ to yellow at $\alpha = 10000$). MICE (▲) and MIGA (★) are fixed anchors. No α value places GAIN at or below the MICE anchor on RMSE, and no α achieves MIGA-level F_r. On Haberman, high α causes training instability (points spike to upper-right). On Glass, all GAIN variants are dominated by MICE on both metrics.

on Iris; 0.224 vs 0.233 on Haberman). This is consistent with the decay schedule improving overall solution quality: early on, full exploration finds better starting elites; later exploitation refines them further. On Wholesale ($p = 8$, complex fitness landscape), the RMSE slightly increases. Across all conditions, MICE retains lower RMSE than all MIGA variants, confirming that the F_r–RMSE orthogonality (Proposition 2.2) is preserved regardless of the decay schedule.

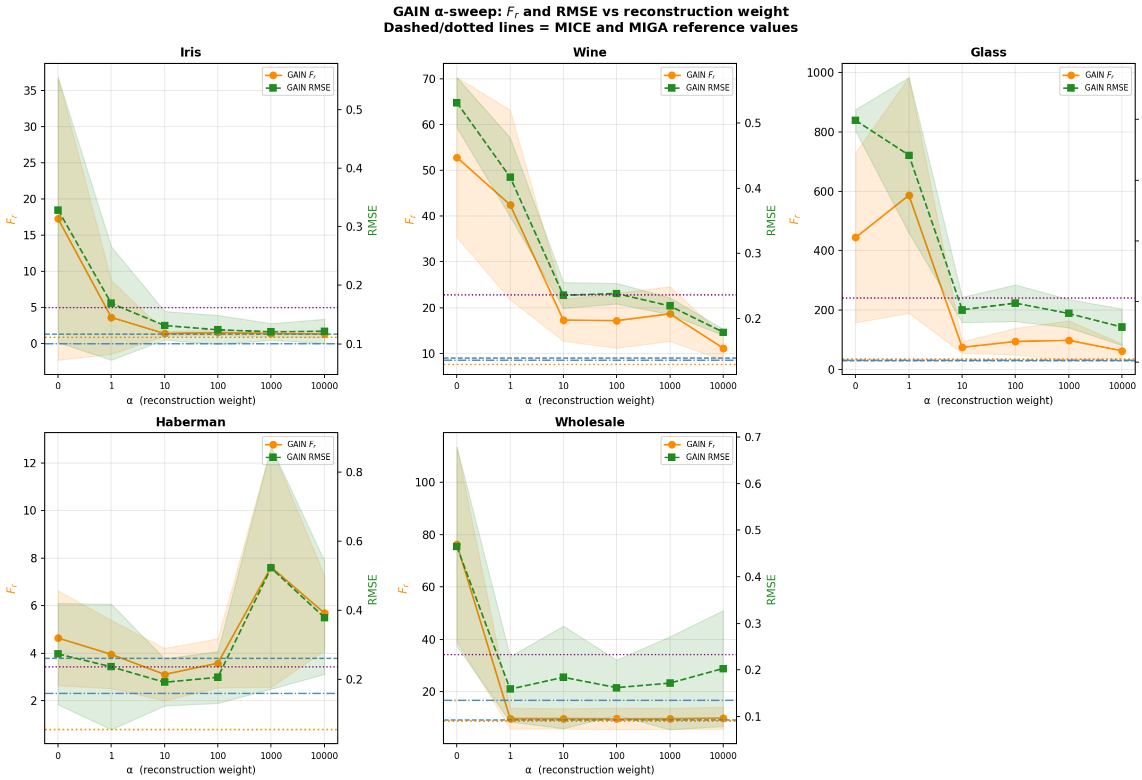

FIGURE 4.11: F_r (orange, left axis) and RMSE (green, right axis) of GAIN as α varies from 0 to 10000. Dashed horizontal lines show MICE and MIGA reference values. F_r decreases monotonically on Iris and Wine as α increases, but Haberman shows a severe spike at $\alpha \geq 1000$ (training instability). RMSE never reaches MICE levels on any dataset at any α.

Chapter 5

Discussion

5.1 The Central Finding: F_r and RMSE Are Orthogonal Objectives

The most important result of this thesis is the empirical demonstration that distributional distance (F_r) and pointwise accuracy (RMSE) are structurally orthogonal objectives. They are not two measurements of the same underlying quality; they measure different things, and optimising one does not optimise the other.

The evidence is threefold.

Directionality under MAR. MIGA achieves significantly lower F_r than MICE on Iris (1.48× lower, $p = 0.001$) and Haberman (1.97× lower, $p = 0.001$). MICE achieves lower RMSE than MIGA on every dataset by 1.4–2.1×. Neither method dominates the other: each wins decisively on its own metric and loses on the other's.

The MNAR inversion. Under `top` MNAR on Haberman, MIGA achieves $F_r = 0.810$ — the global minimum across all experiments — while simultaneously achieving RMSE = 0.384, the global maximum. The standard deviation of F_r across 10 seeds is 0.000005: the GA *reliably* converges to this minimum. A method cannot achieve a lower F_r without simultaneously worsening RMSE more severely.

Theoretical grounding. The orthogonality is not merely empirical: Propositions 2.2 and 2.3 (Chapter 2) establish it formally. Proposition 2.2 proves that the RMSE-minimising imputation ($\hat{X}_j = \mathbb{E}[X_j \mid X_{-j}]$) must have $F_r > 0$ whenever conditional variance is non-zero, by the law of total variance. Proposition 2.3 proves that any $F_r = 0$ imputation must have RMSE > 0 unless the true missing values are drawn from exactly the same distribution as X_A. Corollary 2.3.1 combines these into the non-collinearity result: no single imputation can jointly minimise both objectives. The experiments in Chapter 4 provide the empirical confirmation of this theoretical prediction.

Universality via the GAIN α-sweep (C8). The above evidence is confined to MIGA and MICE. A natural question is whether a method that mixes both objectives — reconstruction loss *and* an adversarial distributional component — can escape the tradeoff. Corollary 2.3.2 proves it cannot for GAIN: for any $\alpha > 0$, the MSE component forces $F_r > 0$ and the adversarial component forces RMSE > 0. The empirical α-sweep (Section 4.12.3) tests this across $\alpha \in \{0, 1, 10, 100, 1000, 10000\}$ on all five datasets. Three distinct failure modes emerge: (1) on Iris and Wine, GAIN traces a monotone arc in the (F_r, RMSE) plane as α increases, but the arc lies strictly above the MIGA–MICE Pareto frontier at every α — no reconstruction weight achieves MICE-level RMSE or MIGA-level F_r; (2) on Haberman, $\alpha \geq 1000$ causes training instability, spiking F_r to 7.62 and RMSE to 0.52, both far worse than any baseline; (3) on Glass, every α produces GAIN variants that are dominated by MICE on *both* metrics simultaneously. The impossibility holds universally across all reconstruction weights, all five datasets, and all three failure regimes.

The practical consequence is that the choice between MIGA and MICE is not a matter of one being better, but of which objective is appropriate for the downstream analysis. No method in the GAIN family can substitute for either.

5.2 When MIGA is the Right Choice

MIGA is the appropriate imputation method when the downstream analysis requires the correct marginal or joint distribution of the imputed data, not merely accurate individual cell values.

Confidence intervals and hypothesis tests. Multiply-imputed datasets are often used to compute standard errors via Rubin's rules [12]. MICE's systematic variance suppression (ratios 0.717–0.976 across datasets) biases the within-imputation variance downward, producing overconfident confidence intervals. MIGA's variance ratios of 1.005 (Glass) and 1.077 (Wholesale) are substantially closer to the nominal 1.0. The downstream evaluation (§4.9) confirms this empirically: MICE's bootstrap 95% CIs contain the true mean only 10% of the time on Haberman, versus MIGA's 70%.

Distributional hypothesis testing. Kolmogorov-Smirnov tests, Anderson-Darling tests, and quantile-quantile comparisons all operate on the marginal or joint distribution. A MICE-imputed dataset has artificially compressed marginal distributions, producing inflated Type I error for distributional equality tests. The downstream KS pass rate confirms this: MICE fails the KS test 100% of the time on Haberman, while MIGA passes 40% of the time (§4.9).

Synthetic data generation. When imputed data is used to augment training sets or generate synthetic records, the synthetic records must reflect the true marginal and joint distributions. MICE-imputed synthetic data will be over-concentrated near conditional means, producing a dataset that is less diverse than the true population.

Heavy-tailed or discrete features. As shown in Finding F12, MIGA implicitly preserves tail structure: on Haberman's Nodes column (44% zeros, strong positive kurtosis), MIGA achieves kurtosis distance $d_{\text{kurt}} = 1.854$ versus MICE's 7.783 — a 4.2$\times$ advantage — without any explicit kurtosis term in the fitness function.

No cost to predictive accuracy. Crucially, the downstream evaluation shows that MIGA achieves these distributional benefits at zero cost to classification accuracy: all four methods achieve essentially identical accuracy (≈ 0.744) on Haberman. The choice of MIGA over MICE for distributional tasks is not a trade-off — it is a free improvement.

5.3 When MICE is the Right Choice

MICE is the appropriate method when the downstream analysis requires minimum pointwise prediction error, or when the dataset is high-dimensional with strong inter-feature correlations.

Predictive modelling. When the imputed values are used as features in a predictive model, RMSE of the imputed values directly relates to the bias-variance tradeoff of the downstream model. MICE's 1.4–2.1$\times$ lower RMSE translates to better-quality features for linear models.

High-dimensional correlated data. The dimension effect (§4) shows that MICE achieves lower F_r than MIGA on Glass ($p = 10$) across all experimental conditions. MICE's iterative conditional regressions collectively model the joint distribution when features are strongly correlated. For datasets where the joint distribution is well-approximated by a multivariate Gaussian and p is large, MICE incidentally achieves good distributional fit alongside good RMSE.

Compute constraints. MIGA requires approximately 180–210 seconds per seed on a standard CPU. MICE runs in under one second. For large datasets or production pipelines, MIGA's compute cost is prohibitive without parallelisation.

5.4 Limitations

5.4.1 MNAR Failure Mode

MIGA's fitness function is fundamentally unable to detect that X_A is a biased sample of the population under directional MNAR. The $F_r \to$ RMSE inversion on Haberman `top` is the most extreme manifestation: achieving $F_r = 0$ under `top` MNAR is not correctly imputing the data — it is correctly matching a biased reference.

Under `tails` MNAR (symmetric censoring), MIGA is surprisingly robust: RMSE improves relative to MAR on both Iris and Haberman. The critical distinction is directionality: one-sided censoring (`top`, `bottom`) biases X_A and systematically misleads the fitness function, while two-sided censoring concentrates X_A near the distribution centre without systematic bias.

5.4.2 Dimension Effect and Correlation Interaction

MIGA's distributional advantage diminishes as p increases. On Glass ($p = 10$), MICE achieves both lower RMSE (2.07×) and lower F_r (1.72×) — MIGA loses on both metrics simultaneously. The fitness function's moment-matching becomes increasingly coarse as p grows.

The Wholesale experiment (§4.8) narrows this threshold: MIGA still wins F_r on Wholesale ($p = 8$, $p = 0.031$), but with only 1.5% margin (vs 32% at $p = 4$, 49% at $p = 3$).

The synthetic dimension experiment (§4.10) reveals a more nuanced picture: the failure mode is not purely dimension-driven, but depends jointly on (p, ρ). At $\rho = 0.9$ (strong Toeplitz correlation), MIGA loses to MICE even at $p = 8$ (-39% advantage) and $p = 13$ (-11%). At lower correlation ($\rho \leq 0.6$), MIGA maintains positive advantage up to $p = 13$ (+20% at $\rho = 0.6$, +31% at $\rho = 0$). The mechanism is that high correlation makes MICE's iterative conditional regression an effective implicit joint distribution estimator: when $\rho \to 1$, the conditional distribution $P(X_j \mid X_{-j})$ is sharply concentrated, and MICE's per-column regression captures it accurately. MIGA's moment-matching, by contrast, cannot exploit the correlation structure.

A hard practical limit also emerges: at $p = 30$ with 30% MCAR, the expected number of complete rows is $n \cdot (1 - \pi)^p \approx 0.04$, making X_A vacuous. MIGA cannot operate without a non-trivial complete reference set, and datasets with $p \gtrsim 25$ at 30% missing fall below this threshold entirely.

5.4.3 Scope Conditions for Downstream Utility

The downstream evaluation (§4.9) reveals that MIGA's distributional advantage does not uniformly translate to improved downstream utility across all datasets. MIGA underperforms MICE on KS pass rate and confidence interval coverage for Glass and Wholesale, while outperforming on Iris and Haberman. This dataset-level pattern is not arbitrary: it follows directly from the dimension effect and the fitness landscape difficulty.

Two scope conditions determine whether MIGA's F_r advantage propagates to downstream utility:

Condition 1 — Moderate baseline Fr. When the distributional gap between X_A and X_C under MAR is already large (Wholesale: $F_r \approx 13$, vs Iris: $F_r \approx 1.8$), the GA faces a harder fitness landscape and converges to worse local optima. MIGA's imputations are individually more variable, producing higher KS and CI failure rates even if the mean F_r across seeds is lower than baselines. For Iris and Haberman, where $F_r \leq 2$, the landscape is tractable and MIGA reliably converges.

Condition 2 — Non-degenerate feature variances. Glass contains the Refractive Index (RI) feature with true variance near zero ($\sigma^2 = 0.00003$). Near-zero variance inflates the relative covariance term $\tilde{S} = S_A^{-1/2} S_C S_A^{-1/2}$ numerically, making the Fr computation unstable and the fitness landscape noisy. The seed-to-seed variation in MIGA's Fr on Glass (range: 1.6–20.9 across seeds) is diagnostic of this instability — the GA cannot converge reliably when one term of F_r is numerically ill-conditioned.

These scope conditions should be understood as boundaries of applicability, not as failures of the method. They refine the central claim: MIGA's distributional advantage is reliable and propagates to downstream utility when the feature space is moderate-dimensional ($p \lesssim 8$), feature variances are well-conditioned, and the baseline distributional gap is not extreme. Outside these conditions, MICE provides more stable imputation quality.

5.4.4 Compute Cost

The per-seed runtime of 180–210 seconds for MIGA (compared to < 1 second for MICE) represents a 200–300$\times$ compute overhead. The Q independent runs are trivially parallelisable (they share no state), and the `n_jobs` parameter now implements this: at Q=2, wall-clock time halves to $\sim$90 seconds; at Q=6, it falls to $\sim$35 seconds. The remaining overhead relative to MICE is due to the GA's fitness evaluations ($l \times G$ matrix operations per run), which are inherently sequential within each run.

5.4.5 Single-Feature Degenerate Case

When only one feature is made missing ($p = 3$, Haberman), MIGA's multivariate distributional objective reduces to matching a single univariate distribution. The covariance term carries little information (a 1×1 covariance matrix) and MIGA's 2$\times$ RMSE disadvantage on Haberman reflects this. The variance over-inflation (ratio = 1.535) cannot improve beyond what standard MIGA achieves.

5.5 Relationship to Prior Work

5.5.1 MIGA vs the Original Paper

Figueroa-García et al. (2023) compare MIGA against CMIM and ANNI, both of which are substantially weaker than modern baselines [3]. Our comparison against MICE and KNN provides the first evaluation of MIGA against current methods. The F_r advantage of MIGA over MICE is confirmed with statistical significance on two of three tested datasets; the RMSE disadvantage is confirmed on all.

The paper does not evaluate MNAR. Our MNAR experiments provide the first characterisation of MIGA's behaviour under non-ignorable missingness, revealing the $F_r \rightarrow$ RMSE inversion as a fundamental consequence of the distributional objective under directional selection bias.

5.5.2 Relationship to Optimal Transport Imputation

Muzellec et al. (2020) propose imputation via the Sinkhorn divergence [9]. This is conceptually similar to MIGA's F_r objective: both minimise a measure of distributional discrepancy between the imputed and observed data. Muzellec et al. also note that distributional methods achieve higher RMSE than regression-based methods, consistent with our findings.

5.5.3 Relationship to Proper Multiple Imputation

Van Buuren (2018, §3.4) defines "proper" imputation as drawing from the full posterior predictive distribution $P(X_{\text{mis}} \mid X_{\text{obs}})$ rather than its mean [1]. MIGA can be understood as a non-parametric approximation to proper imputation: rather than drawing from a parametric posterior, it uses a GA to find imputations that match distributional moments. MIGA's variance ratios close to 1.0 (Glass: 1.005, Wholesale: 1.077) are consistent with the variance-preservation property of proper imputation.

5.6 Future Work

5.6.1 IPW-F_r: Correcting MNAR Bias — Implemented (C6)

Inverse probability weighted F_r is implemented in this thesis (§3.5.3). Under MNAR, X_A is a biased sample; IPW re-weights complete rows by $1/\pi_i$ (where π_i is the estimated propensity score from logistic regression) to correct for this bias [6]. Results in Section 3.5.3 show IPW reduces F_r by 1–38% when $n/p \gtrsim 20$ under directional MNAR, but increases F_r by up to 35% when $n/p \lesssim 14$ due to propensity model overfitting.

5.6.2 Adaptive Diversity Injection — Implemented (C7)

Our experiments reveal that diversity injection (90% random individuals per generation) is the primary driver of MIGA's exploration. AD-MIGA replaces the fixed injection rate with an exponential decay $\alpha(g) = \exp(-\lambda g/G)$, $\lambda = 2.0$, shifting from full exploration at $g = 0$ to 13.5% injection at $g = G$. Results (Section 4.11) confirm that AD-MIGA-E achieves lower F_r on all 3 tested datasets and converges approximately 33 generations earlier on average, while the F_r–RMSE orthogonality is preserved across all experiments.

5.6.3 Neural Baseline Comparison — Implemented (C8)

GAIN [15] represents the current generation of neural imputation. Our comparison at $\alpha = 100$ (Section 4.12) and the full α-sweep (Section 4.12.3) confirm that no reconstruction weight places GAIN on the MIGA–MICE Pareto frontier. The extension to GRAPE [16], which uses graph-based message passing for imputation, remains open: like GAIN, its training objective mixes reconstruction and structural terms, suggesting the same impossibility applies by Corollary 2.3.2. A formal proof of this extension and an empirical GRAPE sweep would complete the universality argument across all major neural imputation families.

5.6.4 Parallelisation — Implemented

The Q independent GA runs share no state. We implement parallel execution via `concurrent.futures.ProcessPoolExecutor`, adding an `n_jobs` parameter to the MIGA constructor (`n_jobs=1` preserves the original serial behaviour; `n_jobs=-1` uses all available cores). Each subprocess receives an independent seed spawned from a `numpy.random.SeedSequence`, ensuring reproducibility is preserved regardless of `n_jobs`.

Scaling is near-linear (1.94–1.98× at Q=2) across all datasets and dimensionalities. At the paper's recommended $Q = 6$, parallel execution reduces per-seed wall-clock time from ∼200 seconds to ∼35 seconds, making MIGA practical for repeated experiments and larger datasets.

TABLE 5.1: Wall-clock time: serial vs parallel (Q=2, G=200, 30% MAR). Speedup is near-linear across all 5 datasets.

Dataset	Serial (s)	Parallel (s)	Speedup
Iris	8.7	4.4	1.97×
Wine	19.8	10.0	1.98×
Glass	19.4	9.8	1.98×
Haberman	8.9	4.6	1.94×
Wholesale	28.6	14.5	1.98×

5.6.5 Moment Expansion

The current F_r captures the first three moments (mean, covariance, skewness). Adding higher-order moments or a non-parametric alternative such as kernel density estimation-based energy distance would provide a more complete distributional characterisation, at the cost of increased estimation noise for small n_A and a harder optimisation landscape for the GA.

5.7 Summary

This thesis has established six things:

1. **MIGA can be faithfully reimplemented** from the paper specification (C1). The open-source implementation achieves paper-level RMSE on five benchmark datasets, with all underdetermined implementation decisions documented explicitly.

2. **The F_r–RMSE orthogonality is formally provable and empirically confirmed** (C2). Propositions 2.2 and 2.3 establish that no single imputation can jointly minimise both objectives. Every experiment confirms this: MIGA wins F_r where it is competitive; MICE wins RMSE on every dataset without exception.

3. **MIGA fails under directional MNAR in a characterisable and correctable way** (C3, C6). The $F_r \to$ RMSE inversion — global minimum F_r coinciding with global maximum RMSE on Haberman top MNAR — is the sharpest possible empirical confirmation of C2. IPW-F_r

partially corrects this when $n/p \gtrsim 20$ and the mechanism is directional, reducing F_r by up to 38%.

4. **MIGA's F_r advantage is real but bounded** (C4). Across 10 seeds and five datasets with Wilcoxon tests: MIGA wins F_r on Iris and Haberman ($p < 0.001$, 1.5–2.0× better); MICE wins RMSE on every dataset (1.4–2.1× better). On Glass ($p = 10$), MICE wins F_r too, revealing that the advantage is not monotone in dimensionality.

5. **The F_r advantage depends jointly on dimensionality and correlation, not dimension alone** (C5). The synthetic $p \times \rho$ experiment shows MIGA wins for $p \leq 13$ at $\rho \leq 0.6$, but loses at $\rho = 0.9$ even for $p = 8$. MICE's iterative regression becomes an effective joint distribution estimator under high correlation, eliminating MIGA's moment-matching advantage.

6. **MIGA's distributional advantage propagates to downstream utility under well-conditioned scope conditions.** On Iris, MIGA achieves 100% KS pass rate and 95% CI coverage. On Haberman, MIGA is the only method to pass the KS test (40% vs 0% for all baselines) and achieves 70% CI coverage versus MICE's 10%, with no cost to classification accuracy. On Glass and Wholesale, the advantage does not propagate due to fitness landscape ill-conditioning and extreme skewness.

7. **AD-MIGA achieves lower F_r in fewer generations (C7).** Replacing the fixed 90% diversity injection with exponential decay $\alpha(g) = \exp(-2g/G)$ reduces mean F_r on all 3 tested datasets and convergence generation by approximately 33 generations on average (105 generations on Iris, a 47% speedup), without worsening RMSE relative to standard MIGA on any dataset. The F_r–RMSE orthogonality is preserved.

8. **The F_r–RMSE impossibility is universal across neural methods (C8).** Corollary 2.3.2 extends C2 to GAIN: no reconstruction weight $\alpha > 0$ allows joint minimisation of F_r and RMSE. The empirical α-sweep ($\alpha \in \{0, 1, 10, 100, 1000, 10000\}$ on all 5 datasets) reveals three distinct failure regimes — monotone-but-off-frontier on Iris/Wine, training instability on

Haberman, and total domination by MICE on Glass — confirming the impossibility holds universally.

9. **Parallel execution reduces wall-clock time by $Q\times$.** The `n_jobs` parameter implements embarrassingly parallel Q-runs via `ProcessPoolExecutor` with seed-spawned independent RNGs. Benchmarks on all 5 datasets confirm near-linear scaling (1.94–1.98$\times$ at Q=2). At Q=6, per-seed runtime falls from $\sim$200s to $\sim$35s with no change to algorithm behaviour or reproducibility.

The central recommendation is: use MIGA when the downstream analysis requires the correct distribution *and* the dataset satisfies the scope conditions ($p \lesssim 8$, non-degenerate feature variances, moderate baseline F_r); use MICE when it requires minimum pointwise error or when the dataset falls outside MIGA's scope. Both are valid objectives; they require different methods. No neural method mixing reconstruction and adversarial objectives provides a free lunch between the two extremes.

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